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Inventor(s)	Van Maris; Antonius Jeroen Adriaan et al.

Recombinant yeast expressing rubisco and phosphoribulokinase

Abstract

The invention relates to a recombinant yeast cell, in particular a transgenic yeast cell, functionally expressing one or more recombinant, in particular heterologous, nucleic acid sequences encoding ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) and phosphoribulokinase (PRK). The invention further relates to the use of carbon dioxide as an electron acceptor in a recombinant chemotrophic micro-organism, in particular a eukaryotic micro-organism.

Inventors: **Van Maris; Antonius Jeroen Adriaan (Delft, NL), Pronk; Jacobus Thomas (Delft, NL), Guadalupe Medina; Victor Gabriel (Delft, NL), Wisselink; Hendrik Wouter (Delft, NL)**

Applicant: **DANISCO US INC.** (Palo Alto, CA)

Family ID: **1000008781393**

Assignee: **Danisco US Inc. (Palo Alto, CA)**

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4241184	12/1979	Hou	435/150	C12P 7/04
10093937	12/2017	Van Maris et al.	N/A	N/A
10689670	12/2019	Papapetridis et al.	N/A	N/A
10982195	12/2020	De Waal	N/A	C12P 7/06
11186850	12/2020	Papapetridis	N/A	C12P 7/06
11203741	12/2020	De Waal	N/A	C12Y 101/01072
11274310	12/2021	De Bruijn	N/A	C12P 7/06
11384368	12/2021	De Waal	N/A	C12Y 401/01039
11624057	12/2022	De Waal	435/162	C12Y 207/01028
11667886	12/2022	Papapetridis	435/255.2	C12N 9/0006
2008/0085341	12/2007	Dai et al.	N/A	N/A
2011/0269180	12/2010	Brat et al.	N/A	N/A
2012/0064622	12/2011	Fischer	N/A	N/A
2018/0251773	12/2017	Van Maris et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2958986	12/2014	EP	N/A
3318624	12/2017	EP	N/A
2008028019	12/2007	WO	N/A

2009013159	12/2008	WO	N/A
2009/036095	12/2008	WO	N/A
2009062190	12/2008	WO	N/A
2010057022	12/2009	WO	N/A
2011/006126	12/2010	WO	N/A
2011010923	12/2010	WO	N/A
2014/129898	12/2013	WO	N/A

OTHER PUBLICATIONS

Brenda Information on xylitol dehydrogenase, obtained from https://www.brenda-enzymes.org/enzyme.php?ecno=1.1.1.B19, Mar. 9, 2022, 3 pages (Year: 2022). cited by examiner
McKinlay et al., PNAS 107:116699-11675, 2010 (Year: 2010). cited by examiner
GenBank Database Accession No. AAA99178, Oct. 2011, 2 pages. cited by applicant
UniProt Database Accession No. Q3SFL5, 3 pages (Year: 2013). cited by applicant
UniProt Database Accession No. POA6F7, 2 pages (Year: 2013). cited by applicant
UniProt Database Accession No. POA6G1, 2 pages (Year: 2013). cited by applicant
Milanez et al., Gene 66:55-63, (Year: 1988). cited by applicant
UniProt Database Accession No. P09559, 2 p. (Year: 2013). cited by applicant
Chang et al., J. Biol. Chem. 279:13778-13785, (Year: 2004). cited by applicant
Nissen et al., Met. Engineer. 2:69-77, (Year: 2000). cited by applicant
Letain et al. “Development of a Genetic System for the Chemolithoautotrophic Bacterium Thiobacillus denitrificans”, Appl. Environ. Microbiol. 73:3265-3271, 2007. cited by applicant
Goloubinoff et al., Nature 337:44-47, 1989. cited by applicant
Chang, H., “Mechanisms of De Novo Multi-domain Protein Folding in Bacteria and Eukaryotes”, Dissertation, Ludwig Maximilian University of Munich, 2007. cited by applicant
Mueller-Cajar et al., BioEssays 29:722-724, 2007. cited by applicant
Brandes et al., J. Biol. Chem. 271 :6490-6496, 1996. cited by applicant
English et al., FEMS Microbiol. Lett. 94:111-120, 1992. cited by applicant
D3 Excerpt from GenomeQuest filed on Oct. 21, 2019 in EP Patent Application No. 14708109. cited by applicant
Third Party Observation filed Oct. 31, 2017 in EP Patent Application No. 14708109. cited by applicant
Wikipedia.org entry (Atomosphere of Earth) [downloaded on Jan. 17, 2020]. cited by applicant
Kerner, Michael J. et al., “Proteome-wide Analysis of Chaperonin-Dependent Protein Folding in *Escherichia coli*”, Cell, Jul. 29, 2005, pp. 209-220, vol. 122. cited by applicant
Guo et al., Comp. Struct. Biotechnol. J. 15:161-167, 2017 (Year: 2017). cited by applicant
Zheng Ping et al., “*Escherichia coli* GroE System and GmE-assisted Rubisco Assembly,” Progress in Biotechnology, (2000), vol. 20, No. 3: 18-22. cited by applicant
Torben L. Nissen, et al., “Anaerobic and aerobic batch cultivations of Saccharomyces cerevisiae,” Yeast, (2000), No. 16: 463-474. cited by applicant
Chinese Office Action relating to Chinese Patent Application No. 201480010133.5 dated May 17, 2017. cited by applicant
Goloubinoff, P., et al., Reconstitution of Active Dimeric Ribulose Bisphospate . . . , Nature, vol. 342, No. 6252, pp. 884-889, 1989. cited by applicant
Tabita, F. R., et al., Function, Structure, and Evolution of the Rubisco-Like . . . , Microbiology and Molecular Biology Reviews 200712 US, vol. 71, No. 4, 2007. cited by applicant

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Primary Examiner: Steadman; David

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This patent application is a continuation patent application of U.S. patent application Ser. No. 15/980,413, filed May 15, 2018 is a continuation patent application of U.S. patent application Ser. No. 14/767,661, filed Aug. 13, 2015, now U.S. Pat. No. 10,093,937, issued Oct. 9, 2018, which is a national phase of International Patent Application No. PCT/NL2014/050106, filed Feb. 21, 2014, which claims the benefit of European Patent Application No. 13156448.6, filed Feb. 22, 2013, the disclosures of each of which is incorporated by reference in its entirety.

REFERENCE TO SEQUENCE LISTING SUBMITTED AS A COMPLIANT ASCII TEXT FILE (.txt)
(1) Pursuant to the EFS-Web legal framework and 37 CFR §§ 1.821-825 (see MPEP § 2442.03(a)), a Sequence Listing in the form of an ASCII-compliant text file (entitled "2919208-499002_Sequence_Listing_ST25.txt" created on 18 Nov. 2020, and 66,586 bytes in size) is submitted concurrently with the instant application, and the entire contents of the Sequence Listing are incorporated herein by reference.
FIELD OF THE INVENTION
(2) The invention relates to a recombinant micro-organism having the ability to produce a desired fermentation product, to the functional expression of heterologous peptides in a micro-organism, and to a method for producing a fermentation product wherein said microorganism is used. In a preferred embodiment the micro-organism is a yeast. The invention is further related to a use of CO.sub.2 in micro-organisms.
BACKGROUND OF THE INVENTION
(3) Microbial fermentation processes are applied for industrial production of a broad and rapidly expanding range of chemical compounds from renewable carbohydrate feedstocks.
(4) Especially in anaerobic fermentation processes, redox balancing of the cofactor couple NADH/NAD.sup.+ can cause important constraints on product yields. This challenge is exemplified by the formation of glycerol as major by-product in the industrial production of—for instance—fuel ethanol by *Saccharomyces cerevisiae*, a direct consequence of the need to reoxidize NADH formed in biosynthetic reactions.
(5) Ethanol production by *Saccharomyces cerevisiae* is currently, by volume, the single largest fermentation process in industrial biotechnology, but various other compounds, including other alcohols, carboxylic acids, isoprenoids, amino acids etc, are currently produced in industrial biotechnological processes.
(6) Various approaches have been proposed to improve the fermentative properties of organisms used in industrial biotechnology by genetic modification.
(7) WO 2008/028019 relates to a method for forming fermentation products utilizing a microorganism having at least one heterologous gene sequence, the method comprising the steps of converting at least one carbohydrate to 3-phosphoglycerate and fixing carbon dioxide, wherein at least one of said steps is catalyzed by at least one exogenous enzyme. Further, it relates to a microorganism for forming fermentation products through fermentation of at least one sugar, the microorganism comprising at least one heterologous gene sequence encoding at least one enzyme selected from the group consisting of phosphopentose epimerase, phosphoribulokinase, and ribulose bisphosphate carboxylase.
(8) In an example, a yeast is mentioned wherein a heterologous PRK and a heterologous Rubisco gene are incorporated. In an embodiment the yeast is used for ethanol production. The results (FIG. **24**) show concentrations for transgenic controls and the modified strains. Little difference is noticeable between modified yeast and its corresponding control. No information is apparent

regarding product yield, sugar conversion, yeast growth, evaporation rates of ethanol. Thus, it is apparent that results are not conclusive with respect to an improvement in ethanol yield.
(9) Further, WO 2008/028019 is silent on the problem of glycerol side-product formation.
(10) A major challenge relating to the stoichiometry of yeast-based production of ethanol, but also of other compounds, is that substantial amounts of NADH-dependent side-products (in particular glycerol) are generally formed as a by-product, especially under anaerobic and oxygen-limited conditions or under conditions where respiration is otherwise constrained or absent. It has been estimated that, in typical industrial ethanol processes, up to about 4 wt. % of the sugar feedstock is converted into glycerol (Nissen et al. Yeast 16 (2000) 463-474). Under conditions that are ideal for anaerobic growth, the conversion into glycerol may even be higher, up to about 10%.
(11) Glycerol production under anaerobic conditions is primarily linked to redox metabolism. During anaerobic growth of *S. cerevisiae*, sugar dissimilation occurs via alcoholic fermentation. In this process, the NADH formed in the glycolytic glyceraldehyde-3-phosphate dehydrogenase reaction is reoxidized by converting acetaldehyde, formed by decarboxylation of pyruvate to ethanol via NAD.sup.+-dependent alcohol dehydrogenase. The fixed stoichiometry of this redox-neutral dissimilatory pathway causes problems when a net reduction of NAD.sup.+ to NADH occurs elsewhere in metabolism. Under anaerobic conditions, NADH reoxidation in *S. cerevisiae* is strictly dependent on reduction of sugar to glycerol. Glycerol formation is initiated by reduction of the glycolytic intermediate dihydroxyacetone phosphate (DHAP) to glycerol 3-phosphate (glycerol-3P), a reaction catalyzed by NAD.sup.+-dependent glycerol 3-phosphate dehydrogenase. Subsequently, the glycerol 3-phosphate formed in this reaction is hydrolysed by glycerol-3-phosphatase to yield glycerol and inorganic phosphate. Consequently, glycerol is a major by-product during anaerobic production of ethanol by *S. cerevisiae*, which is undesired as it reduces overall conversion of sugar to ethanol. Further, the presence of glycerol in effluents of ethanol production plants may impose costs for waste-water treatment.
(12) In WO 2011/010923, the NADH-related side-product (glycerol) formation in a process for the production of ethanol from a carbohydrate containing feedstock—in particular a carbohydrate feedstock derived from lignocellulosic biomass-glycerol side-production problem is addressed by providing a recombinant yeast cell comprising one or more recombinant nucleic acid sequences encoding an NAD.sup.+-dependent acetylating acetaldehyde dehydrogenase (EC 1.2.1.10) activity, said cell either lacking enzymatic activity needed for the NADH-dependent glycerol synthesis or the cell having a reduced enzymatic activity with respect to the NADH-dependent glycerol synthesis compared to its corresponding wild-type yeast cell. A cell is described that is effective in essentially eliminating glycerol production. Also, the cell uses acetate to reoxidise NADH, whereby ethanol yield can be increased if an acetate-containing feedstock is used.
(13) Although the described process in WO 2011/010923 is advantageous, there is a continuing need for alternatives, in particular alternatives that also allow the production of a useful organic compound, such as ethanol, without needing acetate or other organic electron acceptor molecules in order to eliminate or at least reduce NADH-dependent side-product synthesis. It would in particular be desirable to provide a microorganism wherein NADH-dependent side-product synthesis is reduced and which allows increased product yield, also in the absence of acetate.
(14) The inventors realised that it may be possible to reduce or even eliminate NADH-dependent side-product synthesis by functionally expressing a recombinant enzyme in a heterotrophic, chemotrophic microorganism cell, in particular a yeast cell, using carbon dioxide as a substrate.
SUMMARY OF THE INVENTION
(15) Accordingly, the present invention relates to the use of carbon dioxide as an electron acceptor in a recombinant chemoheterotrophic micro-organism, in particular a eukaryotic micro-organism. Chemotrophic, (chemo)heterotrophic and autotrophic and other classifications of a microorganism are herein related to the micro-organism before recombination, this organism is herein also referred to as the host. For instance, through recombination as disclosed herein a host micro-organism that

is originally (chemo)heterotroph and not autotrophic may become autotrophic after recombination, since applying what is disclosed herein causes that the recombined organism may assimilate carbon dioxide, thus resulting in (partial) (chemo)autotrophy.
(16) Advantageously, the inventors have found a way to incorporate the carbon dioxide as a co-substrate in metabolic engineering of heterotrophic industrial microorganisms that can be used to improve product yields and/or to reduce side-product formation.
(17) In particular, the inventors found it to be possible to reduce or even eliminate NADH-dependent side-product synthesis by functionally expressing at least two recombinant enzyme from two specific groups in a eukaryotic microorganism, in particular a yeast cell, wherein one of the enzymes catalysis a reaction wherein carbon dioxide is used and the other uses ATP as a cofactor.
(18) Accordingly, the invention further relates to a recombinant, in a particular transgenic, eukaryotic microorganism, in particular a yeast cell, said microorganism functionally expressing one or more recombinant, in particular heterologous, nucleic acid sequences encoding a ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) and a phosphoribulokinase (PRK).

Description

BRIEF DESCRIPTION OF THE DRAWING
(1) FIG. **1** is a graph showing comparative results for Rubisco activity in cell extracts described in Example 5.
DETAILED DESCRIPTION OF THE INVENTION
(2) A microorganism according to the invention has in particular been found advantageous in that in the presence of Rubisco and the PRK NADH-dependent side-product formation (glycerol) is reduced considerably or essentially completely eliminated and production of the desired product can be increased. It is thought that the carbon dioxide acts as an electron acceptor for NADH whereby less NADH is available for the reaction towards the side-product (such as glycerol).
(3) The invention further relates to a method for preparing an organic compound, in particular an alcohol, organic acid or amino acid, comprising converting a carbon source, in particular a carbohydrate or another organic carbon source using a microorganism, thereby forming the organic compound, wherein the microorganism is a microorganism according to the invention or wherein carbon dioxide is used as an electron acceptor in a recombinant chemotrophic or chemoheterotrophic micro-organism.
(4) The invention further relates to a vector for the functional expression of a heterologous polypeptide in a yeast cell, wherein said vector comprises a heterologous nucleic acid sequence encoding Rubisco and PRK, wherein said Rubisco exhibits activity of carbon fixation. The term “a” or “an” as used herein is defined as “at least one” unless specified otherwise.
(5) When referring to a noun (e.g. a compound, an additive, etc.) in the singular, the plural is meant to be included. Thus, when referring to a specific moiety, e.g. “compound”, this means “at least one” of that moiety, e.g. “at least one compound”, unless specified otherwise.
(6) The term ‘or’ as used herein is to be understood as ‘and/or’.
(7) When referring to a compound of which several isomers exist (e.g. a D and an L enantiomer), the compound in principle includes all enantiomers, diastereomers and cis/trans isomers of that compound that may be used in the particular method of the invention; in particular when referring to such as compound, it includes the natural isomer(s).
(8) For the purpose of clarity and a concise description features are described herein as part of the same or separate embodiments, however, it will be appreciated that the scope of the invention may include embodiments having combinations of all or some of the features described”. In view of this passage it is evident to the skilled reader that the variants of claim **1** as filed may be combined with other features described in the application as filed, in particular with features disclosed in the

dependent claims, such claims usually relating to the most preferred embodiments of an invention.
(9) The term ‘fermentation’, ‘fermentative’ and the like is used herein in a classical sense, i.e. to indicate that a process is or has been carried out under anaerobic conditions. Anaerobic conditions are herein defined as conditions without any oxygen or in which essentially no oxygen is consumed by the yeast cell, in particular a yeast cell, and usually corresponds to an oxygen consumption of less than 5 mmol/l.Math.h, in particular to an oxygen consumption of less than 2.5 mmol/l.Math.h, or less than 1 mmol/l.Math.h. More preferably 0 mmol/L/h is consumed (i.e. oxygen consumption is not detectable). This usually corresponds to a dissolved oxygen concentration in the culture broth of less than 5% of air saturation, in particular to a dissolved oxygen concentration of less than 1% of air saturation, or less than 0.2% of air saturation.
(10) The term “yeast” or “yeast cell” refers to a phylogenetically diverse group of single-celled fungi, most of which are in the division of Ascomycota and Basidiomycota. The budding yeasts (“true yeasts”) are classified in the order Saccharomycetales, with *Saccharomyces cerevisiae* as the most well known species.
(11) The term “recombinant (cell)” or “recombinant micro-organism” as used herein, refers to a strain (cell) containing nucleic acid which is the result of one or more genetic modifications using recombinant DNA technique(s) and/or another mutagenic technique(s). In particular a recombinant cell may comprise nucleic acid not present in a corresponding wild-type cell, which nucleic acid has been introduced into that strain (cell) using recombinant DNA techniques (a transgenic cell), or which nucleic acid not present in said wild-type is the result of one or more mutations—for example using recombinant DNA techniques or another mutagenesis technique such as UV-irradiation—in a nucleic acid sequence present in said wild-type (such as a gene encoding a wild-type polypeptide) or wherein the nucleic acid sequence of a gene has been modified to target the polypeptide product (encoding it) towards another cellular compartment. Further, the term “recombinant (cell)” in particular relates to a strain (cell) from which DNA sequences have been removed using recombinant DNA techniques.
(12) The term “transgenic (yeast) cell” as used herein, refers to a strain (cell) containing nucleic acid not naturally occurring in that strain (cell) and which has been introduced into that strain (cell) using recombinant DNA techniques, i.e. a recombinant cell).
(13) The term “mutated” as used herein regarding proteins or polypeptides means that at least one amino acid in the wild-type or naturally occurring protein or polypeptide sequence has been replaced with a different amino acid, inserted or deleted from the sequence via mutagenesis of nucleic acids encoding these amino acids. Mutagenesis is a well-known method in the art, and includes, for example, site-directed mutagenesis by means of PCR or via oligonucleotide-mediated mutagenesis as described in Sambrook et al., Molecular Cloning-A Laboratory Manual, 2nd ed., Vol. 1-3 (1989). The term “mutated” as used herein regarding genes means that at least one nucleotide in the nucleic acid sequence of that gene or a regulatory sequence thereof, has been replaced with a different nucleotide, or has been deleted from the sequence via mutagenesis, resulting in the transcription of a protein sequence with a qualitatively of quantitatively altered function or the knock-out of that gene.
(14) The term “gene”, as used herein, refers to a nucleic acid sequence containing a template for a nucleic acid polymerase, in eukaryotes, RNA polymerase II. Genes are transcribed into mRNAs that are then translated into protein.
(15) The term “nucleic acid” as used herein, includes reference to a deoxyribonucleotide or ribonucleotide polymer, i.e. a polynucleotide, in either single or double-stranded form, and unless otherwise limited, encompasses known analogues having the essential nature of natural nucleotides in that they hybridize to single-stranded nucleic acids in a manner similar to naturally occurring nucleotides (e. g., peptide nucleic acids). A polynucleotide can be full-length or a subsequence of a native or heterologous structural or regulatory gene. Unless otherwise indicated, the term includes reference to the specified sequence as well as the complementary sequence thereof. Thus, DNAs or

RNAs with backbones modified for stability or for other reasons are “polynucleotides” as that term is intended herein. Moreover, DNAs or RNAs comprising unusual bases, such as inosine, or modified bases, such as tritylated bases, to name just two examples, are polynucleotides as the term is used herein. It will be appreciated that a great variety of modifications have been made to DNA and RNA that serve many useful purposes known to those of skill in the art. The term polynucleotide as it is employed herein embraces such chemically, enzymatically or metabolically modified forms of polynucleotides, as well as the chemical forms of DNA and RNA characteristic of viruses and cells, including among other things, simple and complex cells.
(16) The terms “polypeptide”, “peptide” and “protein” are used interchangeably herein to refer to a polymer of amino acid residues. The terms apply to amino acid polymers in which one or more amino acid residue is an artificial chemical analogue of a corresponding naturally occurring amino acid, as well as to naturally occurring amino acid polymers. The essential nature of such analogues of naturally occurring amino acids is that, when incorporated into a protein, that protein is specifically reactive to antibodies elicited to the same protein but consisting entirely of naturally occurring amino acids. The terms “polypeptide”, “peptide” and “protein” are also inclusive of modifications including, but not limited to, glycosylation, lipid attachment, sulphation, gamma-carboxylation of glutamic acid residues, hydroxylation and ADP-ribosylation.
(17) When an enzyme is mentioned with reference to an enzyme class (EC), the enzyme class is a class wherein the enzyme is classified or may be classified, on the basis of the Enzyme Nomenclature provided by the Nomenclature Committee of the International Union of Biochemistry and Molecular Biology (NC-IUBMB), which nomenclature may be found at the NC-IUBMB website chem.qmul.ac.uk/iubmb/enzyme. Other suitable enzymes that have not (yet) been classified in a specified class but may be classified as such, are meant to be included.
(18) If referred herein to a protein or a nucleic acid sequence, such as a gene, by reference to a accession number, this number in particular is used to refer to a protein or nucleic acid sequence (gene) having a sequence as can be found at the National Center for Biotechnology Information (NCBI-NIH) website ncbi.nlm.nih.gov unless specified otherwise.
(19) Every nucleic acid sequence herein that encodes a polypeptide also, by reference to the genetic code, describes every possible silent variation of the nucleic acid. The term “conservatively modified variants” applies to both amino acid and nucleic acid sequences. With respect to particular nucleic acid sequences, conservatively modified variants refers to those nucleic acids which encode identical or conservatively modified variants of the amino acid sequences due to the degeneracy of the genetic code. The term “degeneracy of the genetic code” refers to the fact that a large number of functionally identical nucleic acids encode any given protein. For instance, the codons GCA, GCC, GCG and GCU all encode the amino acid alanine Thus, at every position where an alanine is specified by a codon, the codon can be altered to any of the corresponding codons described without altering the encoded polypeptide. Such nucleic acid variations are “silent variations” and represent one species of conservatively modified variation.
(20) The term “functional homologue” (or in short “homologue”) of a polypeptide having a specific sequence (e.g. SEQ ID NO: X), as used herein, refers to a polypeptide comprising said specific sequence with the proviso that one or more amino acids are substituted, deleted, added, and/or inserted, and which polypeptide has (qualitatively) the same enzymatic functionality for substrate conversion. This functionality may be tested by use of an assay system comprising a recombinant yeast cell comprising an expression vector for the expression of the homologue in yeast, said expression vector comprising a heterologous nucleic acid sequence operably linked to a promoter functional in the yeast and said heterologous nucleic acid sequence encoding the homologous polypeptide of which enzymatic activity for converting acetyl-Coenzyme A to acetaldehyde in the yeast cell is to be tested, and assessing whether said conversion occurs in said cells. Candidate homologues may be identified by using in silico similarity analyses. A detailed example of such an analysis is described in Example 2 of WO2009/013159. The skilled person will be able to derive

there from how suitable candidate homologues may be found and, optionally upon codon (pair) optimization, will be able to test the required functionality of such candidate homologues using a suitable assay system as described above. A suitable homologue represents a polypeptide having an amino acid sequence similar to a specific polypeptide of more than 50%, preferably of 60% or more, in particular of at least 70%, more in particular of at least 80%, at least 90%, at least 95%, at least 97%, at least 98% or at least 99% and having the required enzymatic functionality. With respect to nucleic acid sequences, the term functional homologue is meant to include nucleic acid sequences which differ from another nucleic acid sequence due to the degeneracy of the genetic code and encode the same polypeptide sequence.

(21) Sequence identity is herein defined as a relationship between two or more amino acid (polypeptide or protein) sequences or two or more nucleic acid (polynucleotide) sequences, as determined by comparing the sequences. Usually, sequence identities or similarities are compared over the whole length of the sequences compared. In the art, “identity” also means the degree of sequence relatedness between amino acid or nucleic acid sequences, as the case may be, as determined by the match between strings of such sequences.

(22) Amino acid or nucleotide sequences are said to be homologous when exhibiting a certain level of similarity. Two sequences being homologous indicate a common evolutionary origin. Whether two homologous sequences are closely related or more distantly related is indicated by “percent identity” or “percent similarity”, which is high or low respectively. Although disputed, to indicate “percent identity” or “percent similarity”, “level of homology” or “percent homology” are frequently used interchangeably. A comparison of sequences and determination of percent identity between two sequences can be accomplished using a mathematical algorithm. The skilled person will be aware of the fact that several different computer programs are available to align two sequences and determine the homology between two sequences (Kruskal, J. B. (1983) An overview of sequence comparison In D. Sankoff and J. B. Kruskal, (ed.), Time warps, string edits and macromolecules: the theory and practice of sequence comparison, pp. 1-44 Addison Wesley). The percent identity between two amino acid sequences can be determined using the Needleman and Wunsch algorithm for the alignment of two sequences. (Needleman, S. B. and Wunsch, C. D. (1970) J. Mol. Biol. 48, 443-453). The algorithm aligns amino acid sequences as well as nucleotide sequences. The Needleman-Wunsch algorithm has been implemented in the computer program NEEDLE. For the purpose of this invention the NEEDLE program from the EMBOSS package was used (version 2.8.0 or higher, EMBOSS: The European Molecular Biology Open Software Suite (2000) Rice, P. Longden, I. and Bleasby, A. Trends in Genetics 16, (6) pp 276-277, emboss.bioinformatics.nl). For protein sequences, EBLOSUM62 is used for the substitution matrix. For nucleotide sequences, EDNAFULL is used. Other matrices can be specified. The optional parameters used for alignment of amino acid sequences are a gap-open penalty of 10 and a gap extension penalty of 0.5. The skilled person will appreciate that all these different parameters will yield slightly different results but that the overall percentage identity of two sequences is not significantly altered when using different algorithms.

(23) Global Homology Definition

(24) The homology or identity is the percentage of identical matches between the two full sequences over the total aligned region including any gaps or extensions. The homology or identity between the two aligned sequences is calculated as follows: Number of corresponding positions in the alignment showing an identical amino acid in both sequences divided by the total length of the alignment including the gaps. The identity defined as herein can be obtained from NEEDLE and is labelled in the output of the program as “IDENTITY”.

(25) Longest Identity Definition

(26) The homology or identity between the two aligned sequences is calculated as follows: Number of corresponding positions in the alignment showing an identical amino acid in both sequences divided by the total length of the alignment after subtraction of the total number of gaps in the

alignment. The identity defined as herein can be obtained from NEEDLE by using the NOBRIEF option and is labeled in the output of the program as “longest-identity”.

(27) A variant of a nucleotide or amino acid sequence disclosed herein may also be defined as a nucleotide or amino acid sequence having one or several substitutions, insertions and/or deletions as compared to the nucleotide or amino acid sequence specifically disclosed herein (e.g. in de the sequence listing).

(28) Optionally, in determining the degree of amino acid similarity, the skilled person may also take into account so-called “conservative” amino acid substitutions, as will be clear to the skilled person. Conservative amino acid substitutions refer to the interchangeability of residues having similar side chains. For example, a group of amino acids having aliphatic side chains is glycine, alanine, valine, leucine, and isoleucine; a group of amino acids having aliphatic-hydroxyl side chains is serine and threonine; a group of amino acids having amide-containing side chains is asp aragine and glutamine; a group of amino acids having aromatic side chains is phenylalanine, tyrosine, and tryptophan; a group of amino acids having basic side chains is lysine, arginine, and histidine; and a group of amino acids having sulphur-containing side chains is cysteine and methionine. Preferred conservative amino acids substitution groups are: valine-leucine-isoleucine, phenylalanine-tyrosine, lysine-arginine, alanine-valine, and asparagine-glutamine Substitutional variants of the amino acid sequence disclosed herein are those in which at least one residue in the disclosed sequences has been removed and a different residue inserted in its place. Preferably, the amino acid change is conservative. Preferred conservative substitutions for each of the naturally occurring amino acids are as follows: Ala to ser; Arg to lys; Asn to gln or his; Asp to glu; Cys to ser or ala; Gln to asn; Glu to asp; Gly to pro; His to asn or gln; Ile to leu or val; Leu to ile or val; Lys to arg; gln or glu; Met to leu or ile; Phe to met, leu or tyr; Ser to thr; Thr to ser; Trp to tyr; Tyr to trp or phe; and, Val to ile or leu.

(29) Nucleotide sequences of the invention may also be defined by their capability to hybridise with parts of specific nucleotide sequences disclosed herein, respectively, under moderate, or preferably under stringent hybridisation conditions. Stringent hybridisation conditions are herein defined as conditions that allow a nucleic acid sequence of at least about 25, preferably about 50 nucleotides, 75 or 100 and most preferably of about 200 or more nucleotides, to hybridise at a temperature of about 65° C. in a solution comprising about 1 M salt, preferably 6×SSC or any other solution having a comparable ionic strength, and washing at 65° C. in a solution comprising about 0.1 M salt, or less, preferably 0.2×SSC or any other solution having a comparable ionic strength. Preferably, the hybridisation is performed overnight, i.e. at least for 10 hours and preferably washing is performed for at least one hour with at least two changes of the washing solution. These conditions will usually allow the specific hybridisation of sequences having about 90% or more sequence identity.

(30) Moderate conditions are herein defined as conditions that allow a nucleic acid sequences of at least 50 nucleotides, preferably of about 200 or more nucleotides, to hybridise at a temperature of about 45° C. in a solution comprising about 1 M salt, preferably 6×SSC or any other solution having a comparable ionic strength, and washing at room temperature in a solution comprising about 1 M salt, preferably 6×SSC or any other solution having a comparable ionic strength. Preferably, the hybridisation is performed overnight, i.e. at least for 10 hours, and preferably washing is performed for at least one hour with at least two changes of the washing solution. These conditions will usually allow the specific hybridisation of sequences having up to 50% sequence identity. The person skilled in the art will be able to modify these hybridisation conditions in order to specifically identify sequences varying in identity between 50% and 90%.

(31) “Expression” refers to the transcription of a gene into structural RNA (rRNA, tRNA) or messenger RNA (mRNA) with subsequent translation into a protein.

(32) As used herein, “heterologous” in reference to a nucleic acid or protein is a nucleic acid or protein that originates from a foreign species, or, if from the same species, is substantially modified

from its native form in composition and/or genomic locus by deliberate human intervention. For example, a promoter operably linked to a heterologous structural gene is from a species different from that from which the structural gene was derived, or, if from the same species, one or both are substantially modified from their original form. A heterologous protein may originate from a foreign species or, if from the same species, is substantially modified from its original form by deliberate human intervention.
(33) The term “heterologous expression” refers to the expression of heterologous nucleic acids in a host cell. The expression of heterologous proteins in eukaryotic host cell systems such as yeast are well known to those of skill in the art. A polynucleotide comprising a nucleic acid sequence of a gene encoding an enzyme with a specific activity can be expressed in such a eukaryotic system. In some embodiments, transformed/transfected yeast cells may be employed as expression systems for the expression of the enzymes. Expression of heterologous proteins in yeast is well known. Sherman, F., et al., Methods in Yeast Genetics, Cold Spring Harbor Laboratory (1982) is a well recognized work describing the various methods available to express proteins in yeast. Two widely utilized yeasts are *Saccharomyces cerevisiae* and *Pichia pastoris*. Vectors, strains, and protocols for expression in *Saccharomyces* and *Pichia* are known in the art and available from commercial suppliers (e.g., Invitrogen). Suitable vectors usually have expression control sequences, such as promoters, including 3-phosphoglycerate kinase or alcohol oxidase, and an origin of replication, termination sequences and the like as desired.
(34) As used herein “promoter” is a DNA sequence that directs the transcription of a (structural) gene. Typically, a promoter is located in the 5′-region of a gene, proximal to the transcriptional start site of a (structural) gene. Promoter sequences may be constitutive, inducible or repressible. If a promoter is an inducible promoter, then the rate of transcription increases in response to an inducing agent.
(35) The term “vector” as used herein, includes reference to an autosomal expression vector and to an integration vector used for integration into the chromosome.
(36) The term “expression vector” refers to a DNA molecule, linear or circular, that comprises a segment encoding a polypeptide of interest under the control of (i.e. operably linked to) additional nucleic acid segments that provide for its transcription. Such additional segments may include promoter and terminator sequences, and may optionally include one or more origins of replication, one or more selectable markers, an enhancer, a polyadenylation signal, and the like. Expression vectors are generally derived from plasmid or viral DNA, or may contain elements of both. In particular an expression vector comprises a nucleic acid sequence that comprises in the 5′ to 3′ direction and operably linked: (a) a yeast-recognized transcription and translation initiation region, (b) a coding sequence for a polypeptide of interest, and (c) a yeast-recognized transcription and translation termination region. “Plasmid” refers to autonomously replicating extrachromosomal DNA which is not integrated into a microorganism's genome and is usually circular in nature.
(37) An “integration vector” refers to a DNA molecule, linear or circular, that can be incorporated in a microorganism's genome and provides for stable inheritance of a gene encoding a polypeptide of interest. The integration vector generally comprises one or more segments comprising a gene sequence encoding a polypeptide of interest under the control of (i.e. operably linked to) additional nucleic acid segments that provide for its transcription. Such additional segments may include promoter and terminator sequences, and one or more segments that drive the incorporation of the gene of interest into the genome of the target cell, usually by the process of homologous recombination. Typically, the integration vector will be one which can be transferred into the target cell, but which has a replicon which is nonfunctional in that organism. Integration of the segment comprising the gene of interest may be selected if an appropriate marker is included within that segment.
(38) By “host cell” is meant a cell which contains a vector and supports the replication and/or expression of the vector. Host cells may be prokaryotic cells such as *E. coli*, or eukaryotic cells

such as yeast, insect, amphibian, or mammalian cells. Preferably, host cells are eukaryotic cells of the order of Actinomycetales.
(39) “Transformation” and “transforming”, as used herein, refers to the insertion of an exogenous polynucleotide into a host cell, irrespective of the method used for the insertion, for example, direct uptake, transduction, f-mating or electroporation. The exogenous polynucleotide may be maintained as a non-integrated vector, for example, a plasmid, or alternatively, may be integrated into the host cell genome.
(40) The microorganism, preferably is selected from the group of Saccharomycetaceae, such as *Saccharomyces cerevisiae, Saccharomyces pastorianus, Saccharomyces beticus, Saccharomyces fermentati, Saccharomyces paradoxus, Saccharomyces uvarum* and *Saccharomyces bayanus; Schizosaccharomyces* such as *Schizosaccharomyces pombe, Schizosaccharomyces japonicus, Schizosaccharomyces octosporus* and *Schizosaccharomyces cryophilus; Torulaspora such* as *Torulaspora delbrueckii; Kluyveromyces* such as *Kluyveromyces marxianus; Pichia* such as *Pichia stipitis, Pichia pastoris* or *pichia angusta, Zygosaccharomyces* such as *Zygosaccharomyces bailii; Brettanomyces* such as *Brettanomyces intermedius, Brettanomyces bruxellensis, Brettanomyces anomalus, Brettanomyces custersianus, Brettanomyces naardenensis, Brettanomyces nanus, Dekkera bruxellensis* and *Dekkera anomala; Metschnikowia, Issatchenkia*, such as *Issatchenkia orientalis, Kloeckera* such as *Kloeckera apiculata; Aureobasidium* such as *Aureobasidium pullulans.*
(41) In a highly preferred embodiment, the microorganism is a yeast cell is selected from the group of Saccharomycetaceae. In particular, good results have been achieved with a *Saccharomyces cerevisiae* cell. It has been found possible to use such a cell according to the invention in a method for preparing an alcohol (ethanol) wherein the NADH-dependent side-product formation (glycerol) was reduced by about 90%, and wherein the yield of the desired product (ethanol) was increase by about 10%, compared to a similar cell without Rubisco and PRK.
(42) The Rubisco may in principle be selected from eukaryotic and prokaryotic Rubisco's.
(43) The Rubisco is preferably from a non-phototrophic organism. In particular, the Rubisco may be from a chemolithoautotrophic microorganism.
(44) Good results have been achieved with a bacterial Rubisco. Preferably, the bacterial Rubisco originates from a *Thiobacillus*, in particular, *Thiobacillus denitrificans*, which is chemolithoautotrophic.
(45) The Rubisco may be a single-subunit Rubisco or a Rubisco having more than one subunit. In particular, good results have been achieved with a single-subunit Rubisco.
(46) In particular, good results have been achieved with a form-II Rubisco, more in particular CbbM.
(47) SEQUENCE ID NO: 2 shows the sequence of a particularly preferred Rubisco in accordance with the invention. It is encoded by the cbbM gene from *Thiobacillus denitrificans*. A preferred alternative to this Rubisco, is a functional homologue of this Rubisco, in particular such functional homologue comprising a sequence having at least 80%, 85%, 90% or 95% sequence identity with SEQUENCE ID NO: 2. Suitable natural Rubisco polypeptides are given in Table 1.
(48) TABLE-US-00001 TABLE 1 Rubisco polypeptides Source Accession no. MAX ID (%) *Thiobacillus denitrificans* AAA99178.2 100 *Sideroxydans lithotrophicus* ES-1 YP_003522651.1 94 *Thiothrix nivea* DSM 5205 ZP_10101642.1 91 *Halothiobacillus neapolitanus* c2 YP_003262978.1 90 *Acidithiobacillus ferrooxidans* ATCC YP_002220242.1 88 53993 *Rhodoferax ferrireducens* T118 YP_522655.1 86 *Thiorhodococcus drewsii* AZ1 ZP_08824342.1 85 uncultured prokaryote AGE14067.1 82
(49) In accordance with the invention, the Rubisco is functionally expressed in the microorganism, at least during use in an industrial process for preparing a compound of interest.
(50) To increase the likelihood that herein enzyme activity is expressed at sufficient levels and in active form in the transformed (recombinant) host cells of the invention, the nucleotide sequence

encoding these enzymes, as well as the Rubisco enzyme and other enzymes of the invention (see below), are preferably adapted to optimise their codon usage to that of the host cell in question. The adaptiveness of a nucleotide sequence encoding an enzyme to the codon usage of a host cell may be expressed as codon adaptation index (CAI). The codon adaptation index is herein defined as a measurement of the relative adaptiveness of the codon usage of a gene towards the codon usage of highly expressed genes in a particular host cell or organism. The relative adaptiveness (w) of each codon is the ratio of the usage of each codon, to that of the most abundant codon for the same amino acid. The CAI index is defined as the geometric mean of these relative adaptiveness values. Non-synonymous codons and termination codons (dependent on genetic code) are excluded. CAI values range from 0 to 1, with higher values indicating a higher proportion of the most abundant codons (see Sharp and Li, 1987, Nucleic Acids Research 15: 1281-1295; also see: Jansen et al., 2003, Nucleic Acids Res. 31(8):2242-51). An adapted nucleotide sequence preferably has a CAI of at least 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8 or 0.9. Most preferred are the sequences which have been codon optimised for expression in the fungal host cell in question such as e.g. *S. cerevisiae* cells.
(51) Preferably, the functionally expressed Rubisco has an activity, defined by the rate of ribulose-1,5-bisphosphate-dependent .sup.14C-bicarbonate incorporation by cell extracts of at least 1 nmol.Math.min.sup.−1.Math.(mg protein).sup.−1, in particular an activity of at least 2 nmol.Math.min.sup.−1.Math.(mg protein).sup.−1, more in particular an activity of at least 4 nmol.Math.min.sup.−1.Math.(mg protein).sup.−1. The upper limit for the activity is not critical. In practice, the activity may be about 200 nmol.Math.min.sup.−1.Math.(mg protein).sup.−1 or less, in particular 25 nmol.Math.min.sup.−1.Math.(mg protein).sup.−1, more in particular 15 nmol.Math.min.sup.−1.Math.(mg protein).sup.−1 or less, e.g. about 10 nmol.Math.min.sup.−1.Math.(mg protein).sup.−1 or less. When referred herein to the activity of Rubisco, in particular the activity at 30° C. is meant. The conditions for an assay for determining this Rubisco activity are as found in the Examples, below (Example 4).
(52) A functionally expressed phosphoribulokinase (PRK, (EC 2.7.1.19)) according to the invention is capable of catalysing the chemical reaction:
ATP+D-ribulose 5-phosphate⇄ADP+D-ribulose 1,5-bisphosphate (1)
(53) Thus, the two substrates of this enzyme are ATP and D-ribulose 5-phosphate, whereas its two products are ADP and D-ribulose 1,5-bisphosphate.
(54) PRK belongs to the family of transferases, specifically those transferring phosphorus-containing groups (phosphotransferases) with an alcohol group as acceptor. The systematic name of this enzyme class is ATP:D-ribulose-5-phosphate 1-phosphotransferase. Other names in common use include phosphopentokinase, ribulose-5-phosphate kinase, phosphopentokinase, phosphoribulokinase (phosphorylating), 5-phosphoribulose kinase, ribulose phosphate kinase, PKK, PRuK, and PRK. This enzyme participates in carbon fixation.
(55) The PRK can be from a prokaryote or a eukaryote. Good results have been achieved with a PRK originating from a eukaryote. Preferably the eukaryotic PRK originates from a plant selected from Caryophyllales, in particular from Amaranthaceae, more in particular from *Spinacia.*
(56) As a preferred alternative to PRK from *Spinacia* a functional homologue of PRK from *Spinacia* may be present, in particular a functional homologue comprising a sequence having at least 70%, 75%, 80%. 85%, 90% or 95% sequence identity with SEQUENCE ID NO 4.
(57) Suitable natural PRK polypeptides are given in Table 2.
(58) TABLE-US-00002 TABLE 2 Natural PRK polypeptides suitable for expression Source Accession no. MAX ID (%) *Spinacia oleracea* P09559.1 100 *Medicago truncatula* XP_003612664.1 88 *Arabidopsis thaliana* NP_174486.1 87 *Vitis vinifera* XP_002263724.1 86 *Closterium peracerosum* BAL03266.1 82 *Zea mays* NP_001148258.1 78
In an advantageous embodiment, the recombinant microorganism further comprises a nucleic acid sequence encoding one or more heterologous prokaryotic or eukaryotic molecular chaperones, which—when expressed—are capable of functionally interacting with an enzyme in the

microorganism, in particular with at least one of Rubisco and PRK.
(59) Chaperonins are proteins that provide favourable conditions for the correct folding of other proteins, thus preventing aggregation. Newly made proteins usually must fold from a linear chain of amino acids into a three-dimensional form. Chaperonins belong to a large class of molecules that assist protein folding, called molecular chaperones. The energy to fold proteins is supplied by adenosine triphosphate (ATP). A review article about chaperones that is useful herein is written by Yebenes (2001); “Chaperonins: two rings for folding”; Hugo Yebenes et al. Trends in Biochemical Sciences, August 2011, Vol. 36, No. 8.
(60) In a preferred embodiment, the chaperone or chaperones are from a bacterium, more preferably from *Escherichia*, in particular *E. coli* GroEL and GroEs from *E. coli* may in particular encoded in a microorganism according to the invention. Other preferred chaperones are chaperones from *Saccharomyces*, in particular *Saccharomyces cerevisiae* Hsp10 and Hsp60. If the chaperones are naturally expressed in an organelle such as a mitochondrion (examples are Hsp60 and Hsp10 of *Saccharomyces cerevisiae*) relocation to the cytosol can be achieved e.g. by modifying the native signal sequence of the chaperonins.
(61) In eukaryotes the proteins Hsp60 and Hsp10 are structurally and functionally nearly identical to GroEL and GroES, respectively. Thus, it is contemplated that Hsp60 and Hsp10 from any eukaryotic cell may serve as a chaperone for the Rubisco. See Zeilstra-Ryalls J, Fayet O, Georgopoulos C (1991). “The universally conserved GroE (Hsp60) chaperonins” Annu Rev Microbiol. 45: 301-25. doi:10.1146/annurev.mi.45.100191.001505. PMID 1683763 and Horwich A L, Fenton W A, Chapman E, Farr G W (2007). “Two Families of Chaperonin: Physiology and Mechanism”. Annu Rev Cell Dev Biol. 23: 115-45. doi:10.1146/annurev.cellbio.23.090506.123555. PMID 17489689.
(62) Particularly good results have been achieved with a recombinant yeast cell comprising both the heterologous chaperones GroEL and GroES.
(63) As a preferred alternative to GroEL a functional homologue of GroEL may be present, in particular a functional homologue comprising a sequence having at least 70%, 75%, 80%, 85%, 90% or 95% sequence identity with SEQUENCE ID NO: 10.
(64) Suitable natural chaperones polypeptide homologous to SEQUENCE ID NO: 10 are given in Table 3.
(65) TABLE-US-00003 TABLE 3 Natural chaperones homologous to SEQUENCE ID NO: 10 polypeptides suitable for expression >gi | 115388105 | ref | XP_001211558.1 | :2-101 10 kDa heat shock protein, mitochondrial [*Aspergillus terreus* NIH2624] >gi | 116196854 | ref | XP_001224239.1 | :1-102 conserved hypothetical protein [*Chaetomium globosum* CBS 148.51] >gi | 119175741 | ref | XP_001240050.1 | :3-102 hypothetical protein CIMG_09671 [*Coccidioides immitis* RS] >gi | 119471607 | ref | XP_001258195.1 | :12-111 chaperonin, putative [*Neosartorya fischeri* NRRL181] >gi | 121699818 | ref | XP_001268174.1 | :8-106 chaperonin, putative [*Aspergillus clavatus* NRRL 1] >gi | 126274604 | ref | XP_001387607.1 | :2-102 predicted protein [*Scheffersomyces stipitis* CBS 6054] >gi | 146417701 | ref | XP_001484818.1 | :5-106 conserved hypothetical protein [*Meyerozyma guilliermondii* ATCC 6260] >gi | 154303611 | ref | XP_001552212.1 | :1-102 10 kDa heat shock protein, mitochondrial [*Botryotinia fuckeliana* B05.10] >gi | 156049571 | ref | XP_001590752.1 | :1-102 hypothetical protein SS1G_08492 [*Sclerotinia sclerotiorum* 1980] >gi | 156840987 | ref | XP_001643870.1 | :1-103 hypothetical protein Kpol_495p10 [*Vanderwaltozyma polyspora* DSM 70924] >gi | 169608295 | ref | XP_001797567.1 | :1-101 hypothetical protein SNOG_07218 [*Phaeosphaeria nodorum* SN15] >gi | 171688384 | ref | XP_001909132.1 | :1-102 hypothetical protein [*Podospora anserina* S mat+] >gi | 189189366 | ref | XP_001931022.1 | :71-168 10 kDa chaperonin [*Pyrenophora tritici-repentis* Pt-1C-BFP] >gi | 19075598 | ref | NP_588098.1 | :1-102 mitochondrial heat shock protein Hsp10 (predicted) [*Schizosaccharomyces pombe* 972h-] >gi | 212530240 | ref | XP_002145277.1 | :3-100 chaperonin, putative [*Talaromyces marneffei* ATCC 18224] >gi | 212530242 | ref |

XP_002145278.1 | :3-95 chaperonin, putative [*Talaromyces marneffei* ATCC 18224] >gi | 213404320 | ref | XP_002172932.1 | :1-102 mitochondrial heat shock protein Hsp10 [*Schizosaccharomyces japonicus* yFS275] >gi | 225557301 | gb | EEH05587.1 | :381-478 pre-mRNA polyadenylation factor fip1 [*Ajellomyces capsulatus* G186AR] >gi | 225684092 | gb | EEH22376.1 | :3-100 heat shock protein [*Paracoccidioides brasiliensis* Pb03 >gi | 238490530 | ref | XP_002376502.1 | :2-104 chaperonin, putative [*Aspergillus flavus* NRRL3357 >gi | 238878220 | gb | EEQ41858.1 | :1-106 10 kDa heat shock protein, mitochondrial [*Candida albicans* WO-1] >gi | 240280207 | gb | EER43711.1 | :426-523 pre-mRNA polyadenylation factor fip1 [*Ajellomyces capsulatus* H143] >gi | 241950445 | ref | XP_002417945.1 | :1-103 10 kda chaperonin, putative; 10 kda heat shock protein mitochondrial (hsp10), putative [*Candida dubliniensis* CD36] >gi | 242819222 | ref | XP_002487273.1 | :90-182 chaperonin, putative [*Talaromyces stipitatus* ATC >gi | 254566327 | ref | XP_002490274.1 | :1-102 Putative protein of unknown function [*Komagataella pastoris* GS115] >gi | 254577241 | ref | XP_002494607.1 | :1-103 ZYRO0A05434p [*Zygosaccharomyces rouxii*] >gi | 255717999 | ref | XP_002555280.1 | :1-103 KLTH0G05588p [*Lachancea thermotolerans*] >gi | 255956581 | ref | XP_002569043.1 | :2-101 Pc21g20560 [*Penicillium chrysogenum* Wisconsin 54-1255] >gi | 258572664 | ref | XP_002545094.1 | :16-108 chaperonin GroS [*Uncinocarpus reesii* 1704] >gi | 261190594 | ref | XP_002621706.1 | :3-100 chaperonin [*Ajellomyces dermatitidis* SLH14081] >gi | 295664909 | ref | XP_002793006.1 | :3-100 10 kDa heat shock protein, mitochondrial [*Paracoccidioides* sp. '*lutzii*'Pb01] >gi | 296412657 | ref | XP_002836039.1 | :76-177 hypothetical protein [*Tuber melanosporum* Mel28] >gi | 302307854 | ref | NP_984626.2 | :2-102 AEL235Wp [*Ashbya gossypii* ATCC 10895] >gi | 302894117 | ref | XP_003045939.1 | :1-102 predicted protein [*Nectria haematococca* mpVI 77-13-4] >gi | 303318351 | ref | XP_003069175.1 | :3-100 10 kDa heat shock protein, mitochondrial, putative [*Coccidioides posadasii* C735 delta SOWgp] >gi | 310795300 | gb | EFQ30761.1 | :1-102 chaperonin 10 kDa subunit [*Glomerella graminicola* M1.001] >gi | 315053085 | ref | XP_003175916.1 | :12-109 chaperonin GroS [*Arthroderma gypseum* CBS 118893] >gi | 317032114 | ref | XP_001394060.2 | :334-433 heat shock protein [*Aspergillus niger* CBS 513.88] >gi | 317032116 | ref | XP_001394059.2 | :2-101 heat shock protein [*Aspergillus niger* CBS 513.88] >gi | 320583288 | gb | EFW97503.1 | :6-106 chaperonin, putative heat shock protein, putative [*Ogataea parapolymorpha* DL-1] >gi | 320591507 | gb | EFX03946.1 | :1-102 heat shock protein [*Grosmannia clavigera* kw1407] >gi | 322700925 | gb | EFY92677.1 | :1-102 chaperonin [*Metarhizium acridum* CQMa 102] >gi | 325096696 | gb | EGC50006.1 | :409-506 pre-mRNA polyadenylation factor fip1 [*Ajellomyces capsulatus* H88] >gi | 326471604 | gb | EGD95613.1 | :14-111 chaperonin 10 Kd subunit [*Trichophyton tonsurans* CBS112818] >gi | 327293056 | ref | XP_003231225.1 | :3-100 chaperonin [*Trichophyton rubrum* CBS 118892] >gi | 330942654 | ref | XP_003306155.1 | :37-136 hypothetical protein PTT_19211 [*Pyrenophora teres f. teres* 0-1] >gi | 336268042 | ref | XP_003348786.1 | :47-147 hypothetical protein SMAC_01809 [*Sordaria macrospora khell*] >gi | 340519582 | gb | EGR49820.1 | :1-109 predicted protein [*Trichoderma reesei* QM6a] >gi | 340960105 | gb | EGS21286.1 | :3-103 putative mitochondrial 10 kDa heat shock protein [*Chaetomium thermophilum* var. *thermophilum* DSM 1495] >gi | 342883802 | gb | EGU84224.1 | :1-102 hypothetical protein FOXB_05181 [*Fusarium oxysporum* Fo5176] >gi | 344302342 | gb | EGW32647.1 | :2-102 hypothetical protein SPAPADRAFT_61712 [*Spathaspora passalidarum* NRRL Y-27907] >gi | 345570750 | gb | EGX53571.1 | :1-102 hypothetical protein AOL_s00006g437 [*Arthrobotrys oligospora* ATCC 24927] >gi | 346321154 | gb | EGX90754.1 | :1-102 chaperonin [*Cordyceps militaris* CM01] >gi | 346970393 | gb | EGY13845.1 | :1-102 heat shock protein [*Verticillium dahliae* VdLs.17] >gi | 354548296 | emb | CCE45032.1 | :1-106 hypothetical protein CPAR2_700360 [*Candida parapsilosis*] >gi | 358385052 | gb | EHK22649.1 | :1-102 hypothetical protein TRIVIDRAFT_230640 [*Trichoderma virens* Gv 29-8] >gi | 358393422 | gb | EHK42823.1 | :1-101 hypothetical protein TRIATDRAFT_258186 [*Trichoderma atroviride* IMI 206040] >gi | 361126733 | gb | EHK98722.1 | :1-97 putative 10 kDa heat shock protein, mitochondrial [*Glare lozoyensis* 74030] >gi | 363753862 | ref | XP_003647147.1 | :2-102 hypothetical protein Ecym_5593 [*Eremothecium cymbalariae* DBVPG#7215] >gi | 365758401 | gb | EHN00244.1 | :1-106 Hsp10p [*Saccharomyces cerevisiae* × *Saccharomyces kudriavzevii* VIN7] >gi | 365987664 | ref | XP_003670663.1 | :1-103 hypothetical protein NDAI_0F01010 [*Naumovozyma dairenensis* CBS 421] >gi | 366995125 | ref | XP_003677326.1 | :1-103 hypothetical protein NCAS_0G00860 [*Naumovozyma castellii* CBS 4309] >gi | 366999797 | ref | XP_003684634.1 | :1-103 hypothetical protein TPHA_0C00430 [*Tetrapisispora phaffii* CBS 4417] >gi | 367009030 | ref | XP_003679016.1 | :1-103 hypothetical protein TDEL_0A04730 [*Torulaspora delbruekii*] >gi | 367023138 | ref | XP_003660854.1 | :1-104 hypothetical protein MYCTH_59302 [*Myceliophthora thermophila* ATCC 42464] >gi | 367046344 | ref | XP_003653552.1 | :1-102 hypothetical protein THITE_2116070 [*Thielavia terrestris* NRRL8126] >gi | 378726440 | gb | EHY52899.1 | :9-109 chaperonin GroES [*Exophiala dermatitidis* NIH/UT8656] >gi | 380493977 | emb | CCF33483.1 | :1-102 chaperonin 10 kDa subunit [*Colletotrichum higginsianu* >gi | 385305728 | gb | EIF49680.1 | :1-102 10 kda heat shock mitochondrial [*Dekkera bruxellensis* AWRI1499] >gi | 389628546 | ref | XP_003711926.1 | :1-102 hsp10-like protein [*Magnaporthe oryzae* 70-15] >gi | 396462608 | ref | XP_003835915.1 | :1-101 similar to 10 kDa heat shock protein [*Leptosphaeria maculans* JN3] >gi | 398392541 | ref | XP_003849730.1 | :1-102 hypothetical protein MYCGRDRAFT_105721 [*Zymoseptoria tritici* IPO323] >gi | 400597723 | gb | EJP65453.1 | :24-124 chaperonin 10 kDa subunit [*Beauveria bassiana* ARSEF 2860] >gi | 401623646 | gb | EJS41738.1 | :1-106 hsp10p [*Saccharomyces arboricola* H-6] >gi | 401842164 | gb | EJT44422.1 | :1-92 HSP10-like protein [*Saccharomyces kudriavzevii* IFO 1802] >gi | 402084027 | gb | EJT79045.1 | :1-102 hsp10-like protein [*Gaeumannomyces graminis* var. triti >gi | 403215209 | emb | CCK69709.1 | :1-104 hypothetical protein KNAG_0C06130 [*Kazachstania naganishii* CBS 8797] >gi | 406604629 | emb | CCH43969.1 | :4-100 hypothetical protein BN7_3524 [*Wickerhamomyces ciferrii*] >gi | 406867021 | gb | EKD20060.1 | :56-156 hypothetical protein MBM_02012 [*Marssonina brunnea* f. sp. ‘multigermtubi’ MB_m1] >gi | 407926227 | gb | EKG19196.1 | :74-174 GroES-like protein [*Macrophomina phaseolina* MS6] >gi | 408398157 | gb | EKJ77291.1 | :11-111 hypothetical protein FPSE_02566 [*Fusarium pseudograminearum* CS3096] >gi | 410082063 | ref | XP_003958610.1 | :1-103 hypothetical protein KAFR_0H00660 [*Kazachstania africana* CBS2517] >gi | 425777664 | gb | EKV15823.1 | :58-157 Chaperonin, putative [*Penicillium digitatum* Pd1] >gi | 440639680 | gb | ELR09599.1 | :1-102 chaperonin GroES [*Geomyces destructans* 20631-21] >gi | 444323906 | ref | XP_004182593.1 | :1-105 hypothetical protein TBLA_0J00760 [*Tetrapisisporablattae* CBS 6284] >gi | 448083208 | ref | XP_004195335.1 | :2-101 Piso0_005888 [*Millerozyma farinosa* CBS 7064] >gi | 448087837 | ref | XP_004196425.1 | :2-102 Piso0_005888 [*Millerozyma farinosa* CBS 7064] >gi | 448534948 | ref | XP_003870866.1 | :1-106 Hsp10 protein [*Candida orthopsilosis* Co 90-125] >gi | 449295977 | gb | EMC91998.1 | :1-102 hypothetical protein BAUCODRAFT_39148 [*Baudoinia compn* >gi | 46123659 | ref | XP_386383.1 | :3-103 hypothetical protein FG06207.1 [*Gibberella zeae* PH-1] >gi | 50289455 | ref | XP_447159.1 | :1-103 hypothetical protein [*Candida glabrata* CBS 138] >gi | 50308731 | ref | XP_454370.1 | :1-103 hypothetical protein [*Kluyveromyces lactis* NRRL Y-1140] >gi | 50411066 | ref | XP_457014.1 | :1-106 DEHA2B01122p [*Debaryomyces hansenii* CBS767] >gi | 50545998 | ref | XP_500536.1 | :1-102 YALI0B05610p [*Yarrowia lipolytica*] >gi | 51013895 | gb | AAT93241.1 | :1-106 YOR020C [*Saccharomyces cerevisiae*] >gi | 6324594 | ref | NP_014663.1 | :1-106 Hsp10p [*Saccharomyces cerevisiae* S288c] >gi | 67523953 | ref | XP_660036.1 | :2-101 hypothetical protein AN2432.2 [*Aspergillus nidulans* FGSC A4] >gi | 70992219 | ref | XP_750958.1 | :12-106 chaperonin [*Aspergillus fumigatus* Af293] >gi | 85079266 | ref | XP_956315.1 | :1-104 hypothetical protein NCU04334 [*Neurospora crassa* OR74A]

(66) As a preferred alternative to GroES a functional homologue of GroES may be present, in particular a functional homologue comprising a sequence having at least 70%, 75%, 80%, 85%,

90% or 95% sequence identity with SEQUENCE ID NO: 12.
(67) Suitable natural chaperones polypeptides homologous to SEQUENCE ID NO: 12 are given in Table 4.
(68) TABLE-US-00004 TABLE 4 Natural chaperones homologous to SEQUENCE ID NO: 12 polypeptides suitable for expression >gi | 115443330 | ref | XP_001218472.1 | heat shock protein 60, mitochondrial precursor [*Aspergillus terreus* NIH2624] >gi | 114188341 | gb | EAU30041.1 | heat shock protein 60, mitochondrial precursor [*Aspergillus terreus* NIH2624] >gi | 119480793 | ref | XP_001260425.1 | antigenic mitochondrial protein HSP60, putative [*Neosartorya fischeri* NRRL 181] >gi | 119408579 | gb | EAW18528.1 | antigenic mitochondrial protein HSP60, putative [*Neosartorya fischeri* NRRL 181] >gi | 126138730 | ref | XP_001385888.1 | hypothetical protein PICST_90190 [*Scheffersomyces stipitis* CBS 6054] >gi | 126093166 | gb | ABN67859.1 | mitochondrial groEL-type heat shock protein [*Scheffersomyces stipitis* CBS 6054] >gi | 145246630 | ref | XP_001395564.1 | heat shock protein 60 [*Aspergillus niger* CBS 513.88] >gi | 134080285 | emb | CAK46207.1 | unnamed protein product [*Aspergillus niger*] >gi | 350636909 | gb | EHA25267.1 | hypothetical protein ASPNIDRAFT_54001 [*Aspergillus niger* ATCC 1015] >gi | 146413148 | ref | XP_001482545.1 | heat shock protein 60, mitochondrial precursor [*Meyerozyma guilliermondii* ATCC 6260] >gi | 154277022 | ref | XP_001539356.1 | heat shock protein 60, mitochondrial precursor [*Ajellomyces capsulatus* NAm1] >gi | 150414429 | gb | EDN09794.1 | heat shock protein 60, mitochondrial precursor [*Ajellomyces capsulatus* NAm1] >gi | 154303540 | ref | XP_001552177.1 | heat shock protein 60 [*Botryotinia fuckeliana* B05.10] >gi | 347840915 | emb | CCD55487.1 | similar to heat shock protein 60 [*Botryotinia fuckeliana*] >gi | 156063938 | ref | XP_001597891.1 | heat shock protein 60, mitochondrial precursor [*Sclerotinia sclerotiorum* 1980] >gi | 154697421 | gb | EDN97159.1 | heat shock protein 60, mitochondrial precursor [*Sclerotinia sclerotiorum* 1980 UF-70] >gi | 156844469 | ref | XP_001645297.1 | hypothetical protein Kpol_1037p35 [*Vanderwaltozyma polyspora* DSM 70294] >gi | 156115957 | gb | EDO17439.1 | hypothetical protein Kpol_1037p35 [*Vanderwaltozyma polyspora* DSM 70294] >gi | 16416029 | emb | CAB91379.2 | probable heat-shock protein hsp60 [*Neurospora crassa*] >gi | 350289516 | gb | EGZ70741.1 | putative heat-shock protein hsp60 [*Neurospora tetrasperma* FGSC 2509] >gi | 169626377 | ref | XP_001806589.1 | hypothetical protein SNOG_16475 [*Phaeosphaeria nodorum* SN15] >gi | 111055053 | gb | EAT76173.1 | hypothetical protein SNOG_16475 [*Phaeosphaeria nodorum* SN15] >gi | 169783766 | ref | XP_001826345.1 | heat shock protein 60 [*Aspergillus oryzae* RIB40] >gi | 238493601 | ref | XP_002378037.1 | antigenic mitochondrial protein HSP60, putative [*Aspergillus flavus* NRRL3357] >gi | 83775089 | dbj | BAE65212.1 | unnamed protein product [*Aspergillus oryzae* RIB40] >gi | 220696531 | gb | EED52873.1 | antigenic mitochondrial protein HSP60, putative [*Aspergillus flavus* NRRL3357] >gi | 391869413 | gb | EIT78611.1 | chaperonin, Cpn60/Hsp60p [*Aspergillus oryzae* 3.042] >gi | 189190432 | ref | XP_001931555.1 | heat shock protein 60, mitochondrial precursor [*Pyrenophora tritici-repentis* Pt-1C-BFP] >gi | 187973161 | gb | EDU40660.1 | heat shock protein 60, mitochondrial precursor [*Pyrenophora tritici-repentis* Pt-1C-BFP] >gi | 190348913 | gb | EDK41467.2 | heat shock protein 60, mitochondrial precursor [*Meyerozyma guilliermondii* ATCC 6260] >gi | 225554633 | gb | EEH02929.1 | hsp60-like protein [*Ajellomyces capsulatus* G186AR] >gi | 238880068 | gb | EEQ43706.1 | heat shock protein 60, mitochondrial precursor [*Candida albicans* WO-1] >gi | 239613490 | gb | EEQ90477.1 | chaperonin GroL [*Ajellomyces dermatitidis* ER-3] >gi | 240276977 | gb | EER40487.1 | hsp60-like protein [*Ajellomyces capsulatus* H143] >gi | 241958890 | ref | XP_002422164.1 | heat shock protein 60, mitochondrial precursor, putative [*Candida dubliniensis* CD36] >gi | 223645509 | emb | CAX40168.1 | heat shock protein 60, mitochondrial precursor, putative [*Candida dubliniensis* CD36] >gi | 254572906 | ref | XP_002493562.1 | Tetradecameric mitochondrial chaperonin [*Komagataella pastoris* GS115] >gi | 238033361 | emb | CAY71383.1 | Tetradecameric mitochondrial chaperonin [*Komagataella pastoris* GS115] >gi | 254579947 | ref | XP_002495959.1 | ZYRO0C07106p [*Zygosaccharomyces rouxii*] >gi | 238938850 | emb |

CAR27026.1 | ZYRO0C07106p [*Zygosaccharomyces rouxii*] >gi | 255712781 | ref | XP_002552673.1 | KLTH0C10428p [*Lachancea thermotolerans*] >gi | 238934052 | emb | CAR22235.1 | KLTH0C10428p [*Lachancea thermotolerans* CBS 6340] >gi | 255721795 | ref | XP_002545832.1 | heat shock protein 60, mitochondrial precursor [*Candida tropicalis* MYA-3404] >gi | 240136321 | gb | EER35874.1 | heat shock protein 60, mitochondrial precursor [*Candida tropicalis* MYA-3404] >gi | 255941288 | ref | XP_002561413.1 | Pc16g11070 [*Penicillium chrysogenum* Wisconsin 54-1255] >gi | 211586036 | emb | CAP93777.1 | Pc16g11070 [*Penicillium chrysogenum* Wisconsin 54-1255] >gi | 259148241 | emb | CAY81488.1 | Hsp60p [*Saccharomyces cerevisiae* EC1118] >gi | 260950325 | ref | XP_002619459.1 | heat shock protein 60, mitochondrial precursor [*Clavispora lusitaniae* ATCC 42720] >gi | 238847031 | gb | EEQ36495.1 | heat shock protein 60, mitochondrial precursor [*Clavispora lusitaniae* ATCC 42720] >gi | 261194577 | ref | XP_002623693.1 | chaperonin GroL [*Ajellomyces dermatitidis* SLH14081] >gi | 239588231 | gb | EEQ70874.1 | chaperonin GroL [*Ajellomyces dermatitidis* SLH14081] >gi | 327355067 | gb | EGE83924.1 | chaperonin GroL [*Ajellomyces dermatitidis* ATCC 18188] >gi | 296422271 | ref | XP_002840685.1 | hypothetical protein [*Tuber melanosporum* Mel28] >gi | 295636906 | emb | CAZ84876.1 | unnamed protein product [*Tuber melanosporum*] >gi | 296809035 | ref | XP_002844856.1 | heat shock protein 60 [*Arthroderma otae* CBS 113480] >gi | 238844339 | gb | EEQ34001.1 | heat shock protein 60 [*Arthroderma otae* CBS 113480] >gi | 302308696 | ref | NP_985702.2 | AFR155Wp [*Ashbya gossypii* ATCC 10895] >gi | 299790751 | gb | AAS53526.2 | AFR155Wp [*Ashbya gossypii* ATCC 10895] >gi | 374108933 | gb | AEY97839.1 | FAFR155Wp [*Ashbya gossypii* FDAG1] >gi | 302412525 | ref | XP_003004095.1 | heat shock protein [*Verticillium albo-atrum* VaMs.102] >gi | 261356671 | gb | EEY19099.1 | heat shock protein [*Verticillium albo- atrum* VaMs. 102] >gi | 302505585 | ref | XP_003014499.1 | hypothetical protein ARB_07061 [*Arthroderma benhamiae* CBS 112371] >gi | 291178320 | gb | EFE34110.1 | hypothetical protein ARB_07061 [*Arthroderma benhamiae* CBS 112371] >gi | 302656385 | ref | XP_003019946.1 | hypothetical protein TRV_05992 [*Trichophyton verrucosum* HKI 0517] >gi | 291183723 | gb | EFE39322.1 | hypothetical protein TRV_05992 [*Trichophyton verrucosum* HKI 0517] >gi | 302915513 | ref | XP_003051567.1 | predicted protein [*Nectria haematococca* mpVI 77-13-4] >gi | 256732506 | gb | EEU45854.1 | predicted protein [*Nectria haematococca* mpVI 77-13-4] >gi | 310794550 | gb | EFQ30011.1 | chaperonin GroL [*Glomerella graminicola* M1.001] >gi | 315048491 | ref | XP_003173620.1 | chaperonin GroL [*Arthroderma gypseum* CBS 118893] >gi | 311341587 | gb | EFR00790.1 | chaperonin GroL [*Arthroderma gypseum* CBS 118893] >gi | 320580028 | gb | EFW94251.1 | Tetradecameric mitochondrial chaperonin [*Ogataea parapolymorpha* DL-1] >gi | 320586014 | gb | EFW98693.1 | heat shock protein mitochondrial precursor [*Grosmannia clavigera* kw1407] >gi | 322692465 | gb | EFY84374.1 | Heat shock protein 60 precursor (Antigen HIS-62) [*Metarhizium acridum* CQMa 102] >gi | 322705285 | gb | EFY96872.1 | Heat shock protein 60 (Antigen HIS-62) [*Metarhizium anisopliae* ARSEF 23] >gi | 323303806 | gb | EGA57589.1 | Hsp60p [*Saccharomyces cerevisiae* FostersB] >gi | 323307999 | gb | EGA61254.1 | Hsp60p [*Saccharomyces cerevisiae* FostersO] >gi | 323332364 | gb | EGA73773.1 | Hsp60p [*Saccharomyces cerevisiae* AWRI796] >gi | 326468648 | gb | EGD92657.1 | heat shock protein 60 [*Trichophyton tonsurans* CBS 112818] >gi | 326479866 | gb | EGE03876.1 | chaperonin GroL [*Trichophyton equinum* CBS 127.97] >gi | 330915493 | ref | XP_003297052.1 | hypothetical protein PTT_07333 [*Pyrenophora teres f. teres* 0-1] >gi | 311330479 | gb | EFQ94847.1 | hypothetical protein PTT_07333 [*Pyrenophora teres f. teres* 0-1] >gi | 336271815 | ref | XP_003350665.1 | hypothetical protein SMAC_02337 [*Sordaria macrospora k-hell*] >gi | 380094827 | emb | CCC07329.1 | unnamed protein product [*Sordaria macrospora k-hell*] >gi | 336468236 | gb | EGO56399.1 | hypothetical protein NEUTE1DRAFT_122948 [*Neurospora tetrasperma* FGSC 2508] >gi | 340522598 | gb | EGR52831.1 | hsp60 mitochondrial precursor-like protein [*Trichoderma reesei* QM6a] >gi | 341038907 | gb | EGS23899.1 | mitochondrial heat shock protein 60-like protein [*Chaetomium thermophilum* var. *thermophilum* DSM 1495] >gi | 342886297

| gb | EGU86166.1 | hypothetical protein FOXB_03302 [*Fusarium oxysporum* Fo5176] >gi | 344230084 | gb | EGV61969.1 | chaperonin GroL [*Candida tenuis* ATCC 10573] >gi | 344303739 | gb | EGW33988.1 | hypothetical protein SPAPADRAFT_59397 [*Spathaspora passalidarum* NRRL Y-27907] >gi | 345560428 | gb | EGX43553.1 | hypothetical protein AOL_s00215g289 [*Arthrobotrys oligospora* ATCC 24927] >gi | 346323592 | gb | EGX93190.1 | heat shock protein 60 (Antigen HIS-62) [*Cordyceps militaris* CM01] >gi | 346975286 | gb | EGY18738.1 | heat shock protein [*Verticillium dahliae* VdLs.17] >gi | 354545932 | emb | CCE42661.1 | hypothetical protein CPAR2_203040 [*Candida parapsilosis*] >gi | 358369894 | dbj | GAA86507.1 | heat shock protein 60, mitochondrial precursor [*Aspergillus kawachii* IFO 4308] >gi | 358386867 | gb | EHK24462.1 | hypothetical protein TRIVIDRAFT_79041 [*Trichoderma virens* Gv29-8] >gi | 358399658 | gb | EHK48995.1 | hypothetical protein TRIATDRAFT_297734 [*Trichoderma atroviride* IMI 206040] >gi | 363750488 | ref | XP_003645461.1 | hypothetical protein Ecym_3140 [*Eremothecium cymbalariae* DBVPG#7215] >gi | 356889095 | gb | AET38644.1 | Hypothetical protein Ecym_3140 [*Eremothecium cymbalariae* DBVPG#7215] >gi | 365759369 | gb | EHN01160.1 | Hsp60p [*Saccharomyces cerevisiae* × *Saccharomyces kudriavzevii* VIN7] >gi | 365764091 | gb | EHN05616.1 | Hsp60p [*Saccharomyces cerevisiae* × *Saccharomyces kudriavzevii* VIN7] >gi | 365985626 | ref | XP_003669645.1 | hypothetical protein NDAI_0D00880 [*Naumovozyma dairenensis* CBS 421] >gi | 343768414 | emb | CCD24402.1 | hypothetical protein NDAI_0D00880 [*Naumovozyma dairenensis* CBS 421] >gi | 366995970 | ref | XP_003677748.1 | hypothetical protein NCAS_0H00890 [*Naumovozyma castellii* CBS 4309] >gi | 342303618 | emb | CCC71399.1 | hypothetical protein NCAS_0H00890 [*Naumovozyma castellii* CBS 4309] >gi | 367005154 | ref | XP_003687309.1 | hypothetical protein TPHA_0J00520 [*Tetrapisispora phaffii* CBS 4417] >gi | 357525613 | emb | CCE64875.1 | hypothetical protein TPHA_0J00520 [*Tetrapisispora phaffii* CBS 4417] >gi | 367017005 | ref | XP_003683001.1 | hypothetical protein TDEL_0G04230 [*Torulaspora delbrueckii*] >gi | 359750664 | emb | CCE93790.1 | hypothetical protein TDEL_0G04230 [*Torulaspora delbrueckii*] >gi | 367035486 | ref | XP_003667025.1 | hypothetical protein MYCTH_2097570 [*Myceliophthora thermophila* ATCC 42464] >gi | 347014298 | gb | AEO61780.1 | hypothetical protein MYCTH_2097570 [*Myceliophthora thermophila* ATCC 42464] >gi | 367055018 | ref | XP_003657887.1 | hypothetical protein THITE_127923 [*Thielavia terrestris* NRRL 8126] >gi | 347005153 | gb | AEO71551.1 | hypothetical protein THITE_127923 [*Thielavia terrestris* NRRL 8126] >gi | 378728414 | gb | EHY54873.1 | heat shock protein 60 [*Exophiala dermatitidis* NIH/UT8656] >gi | 380494593 | emb | CCF33032.1 | heat shock protein 60 [*Colletotrichum higginsianum*] >gi | 385305893 | gb | EIF49836.1 | heat shock protein 60 [*Dekkera bruxellensis* AWRI1499] >gi | 389638386 | ref | XP_003716826.1 | heat shock protein 60 [*Magnaporthe oryzae* 70-15] >gi | 351642645 | gb | EHA50507.1 | heat shock protein 60 [*Magnaporthe oryzae* 70-15] >gi | 440474658 | gb | ELQ43388.1 | heat shock protein 60 [*Magnaporthe oryzae* Y34] >gi | 440480475 | gb | ELQ61135.1 | heat shock protein 60 [*Magnaporthe oryzae* P131] >gi | 393243142 | gb | EJD50658.1 | chaperonin GroL [*Auricularia delicata* TFB-10046 SS5] >gi | 396494741 | ref | XP_003844378.1 | similar to heat shock protein 60 [*Leptosphaeria maculans* JN3] >gi | 312220958 | emb | CBY00899.1 | similar to heat shock protein 60 [*Leptosphaeria maculans* JN3] >gi | 398393428 | ref | XP_003850173.1 | chaperone ATPase HSP60 [*Zymoseptoria tritici* IPO323] >gi | 339470051 | gb | EGP85149.1 | hypothetical protein MYCGRDRAFT_75170 [*Zymoseptoria tritici* IPO323] >gi | 401624479 | gb | EJS42535.1 | hsp60p [*Saccharomyces arboricola* H-6] >gi | 401842294 | gb | EJT44530.1 | HSP60-like protein [*Saccharomyces kudriavzevii* IFO 1802] >gi | 402076594 | gb | EJT72017.1 | heat shock protein 60 [*Gaeumannomyces graminis* var. *tritici* R3-111a-1] >gi | 403213867 | emb | CCK68369.1 | hypothetical protein KNAG_0A07160 [*Kazachstania naganishii* CBS 8797] >gi | 406606041 | emb | CCH42514.1 | Heat shock protein 60, mitochondrial [*Wickerhamomyces ciferrii*] >gi | 406863285 | gb | EKD16333.1 | heat shock protein 60 [*Marssonina brunnea* f. sp. ‘multigermtubi’ MB_m1] >gi | 407922985 | gb | EKG16075.1 | Chaperonin Cpn60 [*Macrophomina phaseolina* MS6] >gi |

408399723 | gb | EKJ78816.1 | hypothetical protein FPSE_00959 [*Fusarium pseudograminearum* CS3096] >gi | 410083028 | ref | XP_003959092.1 | hypothetical protein KAFR_0I01760 [*Kazachstania africana* CBS 2517] >gi | 372465682 | emb | CCF59957.1 | hypothetical protein KAFR_0I01760 [*Kazachstania africana* CBS 2517] >gi | 444315528 | ref | XP_004178421.1 | hypothetical protein TBLA_0B00580 [*Tetrapisispora blattae* CBS 6284] >gi | 387511461 | emb | CCH58902.1 | hypothetical protein TBLA_0B00580 [*Tetrapisispora blattae* CBS 6284] >gi | 448090588 | ref | XP_004197110.1 | Piso0_004347 [*Millerozyma farinosa* CBS 7064] >gi | 448095015 | ref | XP_004198141.1 | Piso0_004347 [*Millerozyma farinosa* CBS 7064] >gi | 359378532 | emb | CCE84791.1 | Piso0_004347 [*Millerozyma farinosa* CBS 7064] >gi | 359379563 | emb | CCE83760.1 | Piso0_004347 [*Millerozyma farinosa* CBS 7064] >gi | 448526196 | ref | XP_003869293.1 | Hsp60 heat shock protein [*Candida orthopsilosis* Co 90-125] >gi | 380353646 | emb | CCG23157.1 | Hsp60 heat shock protein [*Candida orthopsilosis*] >gi | 46123737 | ref | XP_386422.1 | HS60_AJECA Heat shock protein 60, mitochondrial precursor (Antigen HIS-62) [*Gibberella zeae* PH-1] >gi | 50292099 | ref | XP_448482.1 | hypothetical protein [*Candida glabrata* CBS 138] >gi | 49527794 | emb | CAG61443.1 | unnamed protein product [*Candida glabrata*] >gi | 50310975 | ref | XP_455510.1 | hypothetical protein [*Kluyveromyces lactis* NRRL Y-1140] >gi | 49644646 | emb | CAG98218.1 | KLLA0F09449p [*Kluyveromyces lactis*] >gi | 50422027 | ref | XP_459575.1 | DEHA2E05808p [*Debaryomyces hansenii* CBS767] >gi | 49655243 | emb | CAG87802.1 | DEHA2E05808p [*Debaryomyces hansenii* CBS767] >gi | 50555023 | ref | XP_504920.1 | YALI0F02805p [*Yarrowia lipolytica*] >gi | 49650790 | emb | CAG77725.1 | YALI0F02805p [*Yarrowia lipolytica* CLIB122] >gi | 6323288 | ref | NP_013360.1 | Hsp60p [*Saccharomyces cerevisiae* S288c] >gi | 123579 | sp | P19882.1 | HSP60_YEAST RecName: Full = Heat shock protein 60, mitochondrial; AltName: Full = CPN60; AltName: Full = P66; AltName: Full = Stimulator factor I 66 kDa component; Flags:Precursor >gi | 171720 | gb | AAA34690.1 | heat shock protein 60 (HSP60) [*Saccharomyces cerevisiae*] >gi | 577181 | gb | AAB67380.1 | Hsp60p: Heat shock protein 60 [*Saccharomyces cerevisiae*] >gi | 151941093 | gb | EDN59473.1 | chaperonin [*Saccharomyces cerevisiae* YJM789] >gi | 190405319 | gb | EDV08586.1 | chaperonin [*Saccharomyces cerevisiae* RM11-1a] >gi | 207342889 | gb | EDZ70518.1 | YLR259Cp- like protein [*Saccharomyces cerevisiae* AWRI1631] >gi | 256271752 | gb | EEU06789.1 | Hsp60p [*Saccharomyces cerevisiae* JAY291] >gi | 285813676 | tpg | DAA09572.1 | TPA: chaperone ATPase HSP60 [*Saccharomyces cerevisiae* S288c] >gi | 323353818 | gb | EGA85673.1 | Hsp60p [*Saccharomyces cerevisiae* VL3] >gi | 349579966 | dbj | GAA25127.1 | K7_Hsp60p [*Saccharomyces cerevisiae* Kyokai no. 7] >gi | 392297765 | gb | EIW08864.1 | Hsp60p [*Saccharomyces cerevisiae* CEN.PK113-7D] >gi | 226279 | prf | | 1504305A mitochondrial assembly factor >gi | 68485963 | ref | XP_713100.1 | heat shock protein 60 [*Candida albicans* SC5314] >gi | 68486010 | ref | XP_713077.1 | heat shock protein 60 [*Candida albicans* SC5314] >gi | 6016258 | sp | O74261.1 | HSP60_CANAL RecName: Full = Heat shock protein 60, mitochondrial; AltName: Full = 60 kDa chaperonin; AltName: Full = Protein Cpn60; Flags: Precursor >gi | 3552009 | gb | AAC34885.1 | heat shock protein 60 [*Candida albicans*] >gi | 46434552 | gb | EAK93958.1 | heat shock protein 60 [*Candida albicans* SC5314] >gi | 46434577 | gb | EAK93982.1 | heat shock protein 60 [*Candida albicans* SC5314] >gi | 71001164 | ref | XP_755263.1 | antigenic mitochondrial protein HSP60 [*Aspergillus fumigatus* Af293] >gi | 66852901 | gb | EAL93225.1 | antigenic mitochondrial protein HSP60, putative [*Aspergillus fumigatus* Af293] >gi | 159129345 | gb | EDP54459.1 | antigenic mitochondrial protein HSP60, putative [*Aspergillus fumigatus* A1163] >gi | 90970323 | gb | ABE02805.1 | heat shock protein 60 [*Rhizophagus intraradices*]

(69) In an embodiment, a 10 kDa chaperone from Table 3 is combined with a matching 60 kDa chaperone from table 4 of the same organism genus or species for expression in the host.

(70) For instance: >gi|189189366|ref|XP_001931022.1|:71-168 10 kDa chaperonin [Pyrenophora tritici-repentis] expressed together with matching >gi|189190432|ref|XP_001931555.1| heat shock

protein 60, mitochondrial precursor [Pyrenophora tritici-repentis Pt-1C-BFP].
(71) All other combinations from Table 3 and 4 similarly made with same organism source are also available to the skilled person for expression.
(72) Further, one may combine a chaperone from Table 3 from one organism with a chaperone from Table 4 from another organism, or one may combine GroES with a chaperone from Table 3, or one may combine GroEL with a chaperone from Table 4.
(73) As follows from the above, the invention further relates to a method for preparing an organic compound comprising converting a carbon source, using a microorganism, thereby forming the organic compound. The method may be carried out under aerobic, oxygen-limited or anaerobic conditions.
(74) The invention allows in particular a reduction in formation of an NADH dependent side-product, especially glycerol, by up to 100%, up to 99%, or up to 90%, compared to said production in a corresponding reference strain. The NADH dependent side-product formation is preferably reduced by more than 10% compared to the corresponding reference strain, in particular by at least 20%, more in particular by at least 50%. NADH dependent side-product production is preferably reduced by 10-100%, in particular by 20-95%, more in particular by 50-90%.
(75) In preferred method wherein Rubisco, or another enzyme capable of catalysing the formation of an organic compound from CO.sub.2 (and another substrate) or another enzyme that catalyses the function of CO.sub.2 as an electron acceptor, is used, the carbon dioxide concentration in the reaction medium is at least 5% of the CO.sub.2 saturation concentration under the reaction conditions, in particular at least 10% of said CO.sub.2 saturation concentration, more in particular at least 20% of said CO.sub.2 saturation concentration. This is in particular advantageous with respect to product yield. The reaction medium may be oversaturated in CO.sub.2 concentration, saturated in CO.sub.2 concentration or may have a concentration below saturation concentration. In a specific embodiment, the CO.sub.2 concentration is 75% of the saturation concentration or less, in particular 50% of said saturation concentration or less, more in particular is 25% of the CO.sub.2 saturation concentration or less.
(76) In a specific embodiment, the carbon dioxide or part thereof is formed in situ by the microorganism. If desired, the method further comprises the step of adding external CO.sub.2 to the reaction system, usually by aeration with CO.sub.2 or a gas mixture containing CO.sub.2, for instance a CO.sub.2/nitrogen mixture. Adding external CO.sub.2 in particular is used to (increase or) maintain the CO.sub.2 within a desired concentration range, if no or insufficient CO.sub.2 is formed in situ.
(77) Determination of the CO.sub.2 concentration in a fluid is within the routine skills of the person skilled in the art. In practice, one may routinely determine the CO.sub.2 concentration in the gas phase above a culture of the yeast (practically the off-gas if the medium is purged with a gas). This can routinely be measured using a commercial gas analyser, such as a RosemountNGA200000 gas analyser (Rosemount Analytical, Orrvile, USA). The concentration in the liquid phase (relative to the saturation concentration), can then be calculated from the measured value in the gas, from the CO.sub.2 saturation concentration and Henri coefficients of under the existing conditions in the method. These parameters are available from handbooks or can be routinely determined.
(78) As a carbon source, in principle any carbon source that the microorganism can use as a substrate can be used. In particular an organic carbon source may be used, selected from the group of carbohydrates and lipids (including fatty acids). Suitable carbohydrates include monosaccharides, disaccharides, and hydrolysed polysaccharides (e.g. hydrolysed starches, lignocellulosic hydrolysates). Although a carboxylic acid may be present, it is not necessary to include a carboxylic acid such as acetic acid, as a carbon source.
(79) It is in particular an advantage of the present invention that an improved ethanol yield and a reduced glycerol production is feasible compared to, e.g., a wild type yeast cell, without needing to intervene in the genome of the cell by inhibition of a glycerol 3-phosphate phosphohydrolase

and/or encoding a glycerol 3-phosphate dehydrogenase gene.

(80) Still, in a specific embodiment, a yeast cell according to the invention may comprise a deletion or disruption of one or more endogenous nucleotide sequence encoding a glycerol 3-phosphate phosphohydrolase and/or encoding a glycerol 3-phosphate dehydrogenase gene:

(81) Herein in the cell, enzymatic activity needed for the NADH-dependent glycerol synthesis is reduced or deleted. The reduction or deleted of this enzymatic activity can be achieved by modifying one or more genes encoding a NAD-dependent glycerol 3-phosphate dehydrogenase activity (GPD) or one or more genes encoding a glycerol phosphate phosphatase activity (GPP), such that the enzyme is expressed considerably less than in the wild-type or such that the gene encoded a polypeptide with reduced activity.

(82) Such modifications can be carried out using commonly known biotechnological techniques, and may in particular include one or more knock-out mutations or site-directed mutagenesis of promoter regions or coding regions of the structural genes encoding GPD and/or GPP. Alternatively, yeast strains that are defective in glycerol production may be obtained by random mutagenesis followed by selection of strains with reduced or absent activity of GPD and/or GPP. *S. cerevisiae* GPD1, GPD2, GPP1 and GPP2 genes are shown in WO 2011/010923, and are disclosed in SEQ ID NO: 24-27 of that application. The contents of this application are incorporated by reference, in particular the contents relating to GPD and/or GPP.

(83) As shown in the Examples below, the invention is in particular found to be advantageous in a process for the production of an alcohol, notably ethanol. However, it is contemplated that the insight that CO.sub.2 can be used as an electron acceptor in microorganisms that do not naturally allow this, has an industrial benefit for other biotechnological processes for the production of organic molecules, in particular organic molecules of a relatively low molecular weight, particularly organic molecules with a molecular weight below 1000 g/mol. The following items are mentioned herein as preferred embodiments of the use of carbon dioxide as an electron acceptor in accordance with the invention. 1. Use of carbon dioxide as an electron acceptor in a recombinant chemotrophic micro-organism is a non-phototrophic eukaryotic micro-organism. 2. Use of carbon dioxide as an electron acceptor in a recombinant chemotrophic micro-organism, wherein the micro-organism produces an organic compound under anaerobic conditions. 3. Use according to item 1 or 2, wherein the carbon dioxide serves as an electron acceptor in a process with NADH as an electron donor. 5. Use according to any of the preceding items, wherein the micro-organism produces an organic compound in a process with an excess production of ATP and/or NADH. 6. Use according to any of the preceding items, wherein the micro-organism comprises a heterologous nucleic acid sequence encoding a polypeptide from a (naturally) autotrophic organism. 7. Use according to item 6, wherein the micro-organism comprises a heterologous nucleic acid sequence encoding a first prokaryotic chaperone for said polypeptide and preferably a nucleic acid sequence encoding a second prokaryotic chaperone—different from the first—for said polypeptide. 8. Use according to item 7, wherein the chaperones are GroEL and GroES. 9. Use according to any of the preceding items, wherein the micro-organism produces an organic compound selected from the group consisting of alcohols (such as methanol, ethanol, propanol, butanol, phenol, polyphenol), ribosomal peptides, antibiotics (such as penicillin), bio-diesel, alkynes, alkenes, isoprenoids, esters, carboxylic acids (such as succinic acid, citric acid, adipic acid, lactic acid), amino acids, polyketides, lipids, and carbohydrates. 10. Use according to any of the preceding items, wherein the microorganism comprises a heterologous nucleic acid sequence functionally expressing a polypeptide selected from the group consisting of carbonic anhydrases, carboxylases, oxygenases, hydrogenases, dehydrogenases, isomerases, aldolases, transketolases, transaldolases, phosphatases, epimerases, kinases, carboxykinases, oxidoreductases, aconitases, fumarases, reductases, lactonases, phosphoenolpyruvate (PEP) carboxylases, phosphoglycerate kinases, glyceraldehyde 3-phosphate dehydrogenases, triose phosphate isomerases, fructose-1,6-bisphosphatases, sedoheptulose-1,7-bisphosphatases, phosphopentose isomerases, phosphopentose epimerase,

phosphoribulokinases (PRK), glucose 6-phosphate dehydrogenases, 6-phosphogluconolactonases, 6-phosphogluconate dehydrogenases, ribulose 5-phosphate isomerases, ribulose 5-phosphate 3-epimerases, Ribulose-1,5-bisphosphate carboxylase oxygenases, lactate dehydrogenases, malate synthases, isocitrate lyases, pyruvate carboxylases, phosphoenolpyruvate carboxykinases, fructose-1,6-bisphosphatases, phosphoglucoisomerases, glucose-6-phosphatases, hexokinases, glucokinases, phosphofructokinases, pyruvate kinases, succinate dehydrogenases, citrate synthases, isocitrate dehydrogenases, α-ketoglutarate dehydrogenases, succinyl-CoA synthetases, malate dehydrogenases, nucleoside-diphosphate kinases, xylose reductases, xylitol dehydrogenases, xylose isomerases, isoprenoid synthases, and xylonate dehydratases. 11. Use according to item 10, wherein the microorganism comprises a heterologous nucleic acid sequence functionally expressing Ribulose-1,5-bisphosphate carboxylase oxygenase (Rubisco) and/or a heterologous nucleic acid sequence functionally expressing a phosphoribulokinase (PRK). 12. Use according to any of the preceding items, wherein the microorganism is selected from the group of is selected from the group consisting of Saccharomycetaceae, *Penicillium, Yarrowia* and *Aspergillus.* 13. Use according to any of the preceding items, wherein the carbon dioxide is used as an electron acceptor to reduce production of an NAD+-dependent side-product or NADH-dependent side-product, such as glycerol, in a process for preparing another organic compound, such as another alcohol or a carboxylic acid. 14. Recombinant micro-organism, in particular a eukaryotic micro-organism, having an enzymatic system allowing the micro-organism to use carbon dioxide as an electron acceptor under chemotrophic (non-phototrophic) conditions, wherein the microorganism is preferably as defined in the prevision items. 15. Recombinant micro-organism according to item 14, wherein the micro-organism has an enzymatic system for producing an organic compound in a process with an excess production of ATP and/or NADH.
(84) The production of the organic compound of interest may take place in a organism known for it usefulness in the production of the organic compound of interest, with the proviso that the organism has been genetically modified to enable the use of carbon dioxide as an electron acceptor in the organism.
(85) Although it is contemplated that the invention is interesting for the production of a variety of industrially relevant organic compounds, a method or use according the invention is in particular considered advantageous for the production of an alcohol, in particular an alcohol selected from the group of ethanol, n-butanol and 2,3-butanediol; or in the production of an organic acid/carboxylate, in particular a carboxylate selected from the group of L-lactate, 3-hydroxypropionate, D-malate, L-malate, succinate, citrate, pyruvate and itaconate.
(86) Regarding the production of ethanol, details are found herein above, when describing the yeast cell comprising PRK and Rubisco and in the examples. The ethanol or another alcohol is preferably produced in a fermentative process.
(87) For the production of several organic acids (carboxylates), e.g. citric acid, an aerobic process is useful. For citric acid production for instance *Aspergillus niger, Yarrowia lipolytica*, or another known citrate producing organism may be used.
(88) An example of an organic acid that is preferably produced anaerobically is lactic acid. Various lactic acid producing bacterial strains and yeast strains that have been engineered for lactate production are generally known in the art.
EXAMPLES
Example 1. Construction of the Expression Vector
(89) Phosphoribulokinase (PRK) cDNA from *Spinacia oleracea* (spinach) (EMBL accession number: X07654.1) was PCR-amplified using Phusion Hot-start polymerase (Finnzymes, Landsmeer, the Netherlands) and the oligonucleotides XbaI_prk-FW2 and RV1_XhoI_prk (Table 5), and was ligated in pCR®-Blunt II-TOPO® (Life Technologies Europe BV, Bleiswijk, the Netherlands).
(90) TABLE-US-00005 TABLE 5 Oligonucleotides SEQ ID NO Name Sequence (5′ to 3′)

Purpose Cloning 12 Xbal_prk_FW2 TGACATCTAGATGTCACAACAACAAACAATTG cloning of PRK into pUDE046. 13 RV1 XhoI prk TGACATCTAGATGTCACAACAACAAACAATTG cloning of PRK into pUDE046. Primers used for in vivo plasmid assembly 14 HR-cbbM-FW-65 TTGTAAAACGACGGCCAGTGAGCGCGCGTAATACG Rubisco cbbM cassette for plasmids ACTCACTATAGGGCGAATTGGGTACAGCTGGAGCT pUDC075, pUDC099, and pUDC100. CAGTTTATCATTATC 15 HR-cbbM-RV-65 GGAATCTGTGTAGTATGCCTGGAATGTCTGCCGTG Rubisco cbbM cassette for plasmids CCATAGCCATGTATGCTGATATGTCGGTACCGGCC pUDC075, pUDC099, and pUDC100 GCAAATTAAAG 16 linker-cbbO2-pRS416 ATCACTCTTACCAGGCTAGGACGACCCTACTCATG Linker fragment for assembly of TATTGAGATCGACGAGATTTCTAGGCCAGCTTTTGT plasmid pUDC099. TCCCTTTAGTGAGGGTTAATTGCGCGCTTGGCGTAA TCATGGTCATAGC 17 linker-cbbM-GroEL GACATATCAGCATACATGGCTATGGCACGGCAGAC Linker fragment for assembly of ATTCCAGGCATACTACACAGATTCCATCACTCTTAC plasmid pUDC100. CAGGCTAGGACGACCCTACTCATGTATTGAGATCG ACGAGATTTCTAGG Primers used for in vivo integration assembly 18 FW pTDH3-HR-CAN1up GTTGGATCCAGTTTTTAATCTGTCGTCAATCGAAAG 1.sup.st cloning expression cassette linker TTTATTTCAGAGTTCTTCAGACTTCTTAACTCCTGT fragment between CAN1 upstream and AAAAACAAAAAAAAAAAAAGGCATAGCAAGCTGG PRK expression cassette (IMI229), and AGCTCAGTTTATC CAN1up-linker and KlLEU2 expression cassette (IMI232). 19 RV linker-iHR2B AGATATACTGCAAAGTCCGGAGCAACAGTCGTATA 1.sup.st cloning fragment: linker fragment ACTCGAGCAGCCCTCTACTTTGTTGTTGCGCTAAGA between CAN1up-linker and PRK GAATGGACC expression cassette (IMI229). 20 RV linker-iHR6 GCTATGACCATGATTACGCCAAGCGCGCAATTAAC 1.sup.st cloning fragment: linker fragment CCTCACTAAAGGGAACAAAAGCTGGTTGCGCTAAG between CAN1up-linker and KlLEU2 AGAATGGACC expression cassette (IMI232). 21 FW pGAL1-prk HR2B CAACAAAGTAGAGGGCTGCTCGAGTTATACGACTG 2.sup.nd cloning fragment: GAL1.sub.p-PRK- TTGCTCCGGACTTTGCAGTATATCTGCTGGAGCTCT CYCl.sub.t expression cassette (IMI229) AGTACGGATT from pUDE046. 22 RV CYC1t-prk HR2 GGAATCTGTGTAGTATGCCTGGAATGTCTGCCGTG 2.sup.nd cloning fragment: GAL1.sub.p-PRK- CCATAGCCATGTATGCTGATATGTCGTACCGGCCG CYC1.sub.t expression cassette (IMI229) CAAATTAAAG from pUDE046. 23 FW HR2-cbbQ2-HR3 GACATATCAGCATACATGGCTATGG 3.sup.rdI cloning fragment: PGI1.sub.p-cbbQ2-TEF2.sub.t cassette (IMI229). 24 RV HR2-cbbQ2-HR3 GGACACGCTTGACAGAATGTCAAAGG 3.sup.rd cloning fragment: PGI1.sub.p-cbbQ2-TEF2.sub.t cassette (IMI229). 25 FW HR3-cbbO2-HR4 CGTCCGATATGATCTGATTGG 4.sup.th TARI cloning fragment: PGK1.sub.p- cbbO2-ADH1.sub.t cassette (IMI229). 26 RV HR3-cbbO2-HR4 CCTAGAAATCTCGTCGATCTC 4.sup.th cloning fragment: PGK1.sub.p-cbbO2- ADH1.sub.t cassette (IMI229). 27 FW HR4-GroEL-HR5 ATCACTCTTACCAGGCTAGG 5.sup.th cloning fragment: TEF1.sub.p-groEL-ACT1.sub.t cassette (IMI229). 28 RV HR4-GroEL-HR5 CTGGACCTTAATCGTGTGCGCATCCTC 5.sup.th cloning fragment: TEF1.sub.p-groEL-ACT1.sub.t cassette (IMI229). 29 FW HRS-GroES-HR6 CCGTATAGCTTAATAGCCAGCTTTATC 6.sup.th cloning fragment: TPI1.sub.p-groES-PGI1.sub.t cassette (IMI229). 30 RV HRS-GroES-HR6 GCTATGACCATGATTACGCCAAGC 6.sup.th cloning fragment: TPI1.sub.p-groES-PGI1.sub.t cassette (IMI229). 31 FW HR6-LEU2-CAN1dwn CCAGCTTTTGTTCCCTTTAGTGAGGGTTAATTGCGC 7.sup.th (IMI229) or 2.sup.nd

(IMI232) cloning GCTTGGCGTAATCATGGTCATAGCCTGTGAAGATC fragment: KlLEU2 cassette from CCAGCAAAG pUG73. 32 RV LEU2 HR-CAN1 AGCTCATTGATCCCTTAAACTTTCTTTTCGGTGTAT 7.sup.th (IMI229) or 2.sup.nd (IMI232) cloning GACTTATGAGGGTGAGAATGCGAAATGGCGTGGA fragment: KlLEU2 cassette from AATGTGATCAAAGGTAATAAAACGTCATATATCCG pUG73. CAGGCTAACCGGAAC Primers used for verification of the in vivo assembled constructs 33 m-PCR-HR1-FW GGCGATTAAGTTGGGTAACG Diagnostic for assembly of plasmids pUDC075, pUDC099, and pUDC100,. 34 m-PCR-HR1-RV AACTGAGCTCCAGCTGTACC Diagnostic for assembly of plasmids pUDC075, pUDC099, pUDC100, and integration in strain IMI229. 35 m-PCR-HR2-FW ACGCGTGTACGCATGTAAC Diagnostic for assembly of pUDC075, pUDC099, pUDC100, and integration in strain IMI229 36 m-PCR-HR2-RV GCGCGTGGCTTCCTATAATC Diagnostic for assembly of pUDC075, pUDC099, pUDC100, and integration in strain IMI229 37 m-PCR-HR3-FW GTGAATGCTGGTCGCTATAC Diagnostic for assembly of pUDC075, pUDC099, pUDC100, and integration in strain IMI229. 38 m-PCR-HR3-RV GTAAGCAGCAACACCTTCAG Diagnostic for assembly of pUDC075, pUDC099, pUDC100, and integration in strain IMI229. 39 m-PCR-HR4-FW ACCTGACCTACAGGAAAGAG Diagnostic for assembly of pUDC075, pUDC099, pUDC100, and integration in strain IMI229. 40 m-PCR-HR4-RV TGAAGTGGTACGGCGATGC Diagnostic for assembly of pUDC075, pUDC099, pUDC100, and integration in strain IMI229. 41 m-PCR-HR5-FW ATAGCCACCCAAGGCATTTC Diagnostic for assembly of pUDC075, pUDC099, pUDC100, and integration in strain IMI229. 42 m-PCR-HR5-RV CCGCACTTTCTCCATGAGG Diagnostic for assembly of pUDC075, pUDC099, pUDC100, and integration in strain IMI229. 43 m-PCR-HR6-FW CGACGGTTACGGTGTTAAG Diagnostic for assembly of pUDC075, pUDC099, pUDC100, and integration in strain IMI229. 44 m-PCR-HR6-RV CTTCCGGCTCCTATGTTGTG Diagnostic for assembly of pUDC075, pUDC099, pUDC100, and integration in strain IMI229.

(91) After restriction by XbaI and XhoI, the PRK-containing fragment was ligated into pTEF424. The TEF1p was later replaced by GAL1p from plasmid pSH47 by XbaI and SacI restriction/ligation, creating plasmid pUDE046 (see Table 6).

(92) TABLE-US-00006 TABLE 6 Plasmids Name Relevant genotype Source/reference pFL451 pAOX1-prk (Spinach)-AOX1t (pHIL2-D2 HIS4 Amp Brandes et al. centromeric) 1996..sup.14 pCR ®-Blunt bla Life II-TOPO Technologies Europe BV pTEF424_TEF TRP1 2μ bla Mumberg et al. 1995.sup.25. pSH47 URA3 CEN6 ARS4 GAL1.sub.p-cre-CYC1.sub.t bla Güldener et al 1996.sup.26 pUD0E46 TRP1 2μ GAL1p-prk-CYC1.sub.t bla This study. pPCR-Script bla Life Technologies Europe BV pGPD_426 URA3 2μ bla Mumberg et al. 1995.sup.25. pRS416 URA3 CEN6 ARS4 bla Mumberg et al. 1995.sup.25. pBTWW002 URA3 2μ TDH3.sub.p-cbbM-CYC1.sub.t bla This study. pUDC098 URA3 CEN6 ARS4 TDH3.sub.p-cbbM-CYC1.sub.t bla This study. pMK-RQ nptII Life Technologies Europe BV pUD230 PGI1.sub.p-cbbQ2-TEF2.sub.t nptII Life Technologies Europe BV pUD231 PGK1.sub.p-cbbO2-ADH1.sub.t nptII Life Technologies Europe BV pUD232 TEF1.sub.p-groEL-ACT1.sub.t nptII Life Technologies Europe BV pUD233 TPI1.sub.p-groES-PGI1.sub.t nptII Life Technologies Europe BV pUDC075 URA3 CEN6 ARS4 TDH3.sub.p-cbbM-CYC1.sub.t;PGI1.sub.p-cbbQ2- This study. TEF2.sub.t;PGK1.sub.p-cbbO2-ADH1.sub.t;TEF1.sub.p-groEL-ACT1.sub.t;TPI1.sub.p- groES-PGI1.sub.t bla pUDC099 URA3 CEN6 ARS4 TDH3.sub.p-cbbM-CYC1.sub.t;PGI1.sub.p-cbbQ2- This study. TEF2.sub.t;PGK1.sub.p-cbbO2-ADH1.sub.t bla pUDC100 URA3 CEN6 ARS4 TDH3.sub.p-cbbM-CYC1.sub.t; TEF1.sub.p-groEL- This study. ACT1.sub.t;TPI1.sub.p-groES-PGI1.sub.t bla

(93) Rubisco form II gene cbbM from *Thiobacillus denitrificans* (*T. denitrificans*) flanked by KpnI and SacI sites was codon optimized synthesized at GeneArt (Life Technologies Europe BV), and ligated into pPCR-Script., the plasmid was then digested by BamHI and SacI. The cbbM-containing fragment was ligated into the BamHI and SacI restricted vector pGPD_426 creating plasmid pBTWW002. The cbbM expression cassette was transferred into pRS416 using KpnI and SacI, yielding pUDC098.
(94) Expression cassette of the specific Rubisco form II cheparones from *T. denitrificans* cbbQ2 and ebbO2, and chaperones groEL and groES from *E. coli*, were condon optimized. The expression cassettes contained a yeast constitutive promoters and terminator, flanking the codon optimized gene. The cassette was flanked by unique 60 bp regions obtained by randomly combining bar-code sequences used in the *Saccharomyces* Genome Deletion Project and an EcoRV site (GeneArt). The expression cassettes were inserted in plasmid pMK-RQ (GeneArt) using the SfiI cloning sites yielding pUB230 (PGI1p-cbbQ2-TEF2t), pUD231 (PGK1p-cbbO2-ADH1t), pUD232 (TEF1p-groEL-ACT1t), and pUDE233 (TPI1p-groES-PGI1t) Table 6). The expression cassette TDH3p-cbbM-CYC1t was PCR-amplified from plasmid pBTWW002 using Phusion Hot-Start Polymerase (Finnzymes) and primers HR-cbbM-FW-65 and HR-cbbM-RV-65 in order to incorporate the 60-bp region for recombination cloning.
Example 2. Strain Construction, Isolation and Maintenance
(95) All *Saccharomyces cerevisiae* strains used (Table 7) belong to the CEN.PK family. All strains were grown in 2% w/v glucose synthetic media supplemented with 150 mg L.sup.−1 uracil when required until they reached end exponential phase, then sterile glycerol was added up to ca. 30% v/v and aliquots of 1 ml were stored at −80° C.
(96) TABLE-US-00007 TABLE 7 *Saccharomyces cerevisiae* strains Strain Relevant genotype Source/reference CEN.PK113-5D MATa ura3-52 Euroscarf CEN.PK102-3A MATa ura3-52 leu2-3, 112 Euroscarf IMC014 MATa ura3-52 pUDC075 (CEN6 ARS4 URA3 TDH3.sub.p- This study. cbbM-CYC1.sub.t PGI1.sub.p-cbbQ2-TEF2.sub.t PGK1.sub.p-cbbO2- ADH1.sub.t TEF1.sub.p-groEL-ACT1.sub.t TPI1.sub.p-groES-PGI1.sub.t) IMC033 MATa ura3-52 pUDC098 (CEN6 ARS4 URA3 TDH3.sub.p- This study. cbbM-CYC1.sub.t) IMC034 MATa ura3-52 pUDC099 (CEN6 ARS4 URA3 TDH3.sub.p- This study. cbbM-CYC1.sub.t PGI1.sub.p-cbbQ2-TEF2.sub.t PGK1.sub.p-cbbO2- ADH1.sub.tcbbO2-pRS416 linker) IMC035 MATa ura3-52 pUDC100 (CEN6 ARS4 URA3 TEF1.sub.p- This study. groEL-ACT1.sub.t TPI1.sub.p-groES-PGI1.sub.t cbbM-GroEL linker) IMI229 MATa ura3-52 leu2-3, 112 can1Δ::GAL1.sub.p-prk-CYC1.sub.t This study. PGI1.sub.p-cbbQ2-TEF2.sub.t,PGK1.sub.p-cbbO2-ADH1.sub.t,TEF1.sub.p- groEL-ACT1.sub.t,TPI1.sub.p-groES-PGI1.sub.t KlLEU2 IMI232 MATa ura3-52 leu2-3, 112 can1Δ::KlLEU2 This study. IMU032 IMI232 p426_GPD (2μ URA3) This study. IMU033 IMI229 pUDC100 (CEN6 ARS4 URA3 TEF1.sub.p-groEL- This study. ACT1.sub.t TPI1.sub.p-groES-PGI1.sub.t cbbM-GroEL linker)
(97) The strain IMC014 that co-expressed the Rubisco form II ccbM and the four chaperones cbbQ2, ebbO2, groEL, and groES was constructed using in vivo transformation associated recombination. 200 fmol of each expression cassette were pooled with 100 fmol of the KpnI/SacI linearized pRS416 backbone in a final volume of 50 μl and transformed in CEN.PK 113-5D using the lithium acetate protocol (Gietz, et al., Yeast Transformation by the LiAc/SS Carrier DNA/PEG Method in Yeast Protocol, *Humana press*, 2006). Cells were selected on synthetic medium. Correct assembly of the fragment of pUDC075 was performed by multiplex PCR on transformant colonies using primers enabling amplification over the regions used for homologous recombination (Table 5) and by restriction analysis after re-transformation of the isolated plasmid in *E. coli* DH5a. PUDC075 was sequenced by Next-Generation Sequencing (Illumina, San Diego, California, U.S.A.) (100br reads paired-end, 50 Mb) and assembled with Velvet (Zerbino, et al., Velvet: Algorithms for De Novo Short Read Assembly Using De Bruijn Graphs, *Genome Research*, 2008). The assembled sequence did not contain mutations in any of the assembled expression cassettes.

The strains IMC034 and IMC035 that expressed ccbM/ccbQ2/ccbO2 and ccbM/groEL/groES respectively were constructed using the same in vivo assembly method with the following modification. To construct plasmids pUDC099 and pUDC100, 120 bp cbbO2-pRS416 linker and cbbM-GroEL linker were used to close the assembly respectively (Table 5), 100 fmol of each of complementary 120 bp oligonucleotides were added to the transformation. The strain IMC033 that only expressed the cbbM gene was constructed by transforming CEN.PK113-5D with pUDC098.

(98) To construct the strain IMU033 that co-expressed PRK, ccbM, ccbQ2, ccbO2, GroEL, GroES, the intermediate strain IMI229 was constructed by integrating PRK, the four chaperones and KlLEU2 (Güldener, et al., A second set of loxP marker cassettes for Cre-mediated multiple gene knockouts in budding yeast, Nucleic Acids Research, 2002) at the CAN1 locus by in vivo homologous integration in CEN.PK102-3A. The expression cassettes were PCR amplified using Phusion Hot-Start Polymerase (Finnzymes, Thermo Fisher Scientific Inc. Massachusetts, U.S.A.), the corresponding oligonucleotides and DNA templates (Table 5). Finally, the strain IMI229 was transformed with pUDC100 that carries the Rubisco form II ccbM and the two *E. coli* chaperones groEL and groES.

(99) Strain IMI232 was constructed by transforming CEN.PK102-3A with the KlLEU2 cassette. IMI232 was finally transformed with the plasmid p426GPD to restore prototrophy resulting in the reference strain IMU032.

Example 3. Experimental Set-Up of Chemostat and Batch Experiments

(100) Anaerobic chemostat cultivation was performed essentially as described (Basso, et al., Engineering topology and kinetics of sucrose metabolism in *Saccharomyces cerevisiae* for improved ethanol yield, Metabolic Engineering 13:694-703, 2011), but with 12.5 g 1-1 glucose and 12.5 g 1-1 galactose as the carbon source and where indicated, a mixture of 10% CO.sub.2/90% N.sub.2 replaced pure nitrogen as the sparging gas. Residual glucose and galactose concentrations were determined after rapid quenching (Mashego, et al., Critical evaluation of sampling techniques for residual glucose determination in carbon-limited chemostat culture of *Saccharomyces cerevisiae, Biotechnology and Bioengineering* 83:395-399, 2003) using commercial enzymatic assays for glucose (Boehringer, Mannheim, Germany) and D-galactose (Megazyme, Bray, Ireland). Anaerobic bioreactor batch cultures were grown essentially as described (Guadalupe Medina, et al., Elimination of glycerol production in anaerobic cultures of a *Saccharomyces cerevisiae* strain engineered to use acetic acid as an electron acceptor. *Applied and Environmental Microbiology* 76:190-195, 2010), but with 20 g L.sup.−1 galactose and a sparging gas consisting of 10% CO.sub.2 and 90% N.sub.2. Biomass and metabolite concentrations in batch and chemostat and batch cultures were determined as described by Guadalupe et al. (Guadalupe Medina, et al., Elimination of glycerol production in anaerobic cultures of a *Saccharomyces cerevisiae* strain engineered to use acetic acid as an electron acceptor. Appl. Environ. Microbiol. 76, 190-195, 2010). In calculations of ethanol fluxes and yields, ethanol evaporation was corrected for based on a first-order evaporation rate constant of 0.008 h.sup.−1 in the bioreactor set-ups and under the conditions used in this study.

Example 4. Enzyme Assays for Phosphoribulokinase (PRK) and Rubisco

(101) Cell extracts for analysis of phosphoribulokinase (PRK) activity were prepared as described previously (Abbott, et al., Catalase Overexpression reduces lactic acid-induced oxidative stress in *Saccharomyces cerevisiae, Applied and Environmental Microbiology* 75:2320-2325, 2009). PRK activity was measured at 30° C. by a coupled spectrophotometric assay (MacElroy, et al., Properties of Phosphoribulokinase from *Thiobacillus neapolitanus*, *Journal of Bacteriology* 112:532-538, 1972). Reaction rates were proportional to the amounts of cell extract added. Protein concentrations were determined by the Lowry method (Lowry, et al., Protein measurement with the Folin phenol reagent, *The Journal of Biological Chemistry* 193:265-275, 1951) using bovine serum albumin as a standard.

(102) Cell extracts for Rubisco activity assays were prepared as described in Abbott, D. A. et al.

Catalase overexpression reduces lactic acid-induced oxidative stress in *Saccharomyces cerevisiae*. Appl. Environ. Microbiol. 75:2320-2325, 2009, with two modifications: Tris-HCl (1 mM, pH 8.2) containing 20 mM $MgCl_2.6H_2O$, 5 mM of DTT 5 mM $NaHCO_3$ was used as sonication buffer and Tris-HCl (100 mM, pH 8.2), 20 mM $MgCl_2.6H_2O$ and 5 mM of DTT as freezing buffer. Rubisco activity was determined by measuring $^{14}CO_2$-fixation (PerkinElmer, Groningen, The Netherlands) as described (Beudeker, et al., Relations between d-ribulose-1,5-biphosphate carboxylase, carboxysomes and CO_2 fixing capacity in the obligate chemolithotroph *Thiobacillus neapolitanus* grown under different limitations in the chemostat, Archives of Microbiology 124:185-189, 1980) and measuring radioactive counts in a TRI-CARB® 2700TR Series liquid scintillation counter (PerkinElmer, Groningen, The Netherlands), using Ultima Gold™ scintillation cocktail (PerkinElmer, Groningen, The Netherlands). Protein concentrations were determined by the Lowry method (Lowry, O. H., Rosebrough, N. J., Farr, A. L., & Randall, R. J. Protein measurement with the Folin phenol reagent. J. Biol. Chem. 193:265-275, 1951) using standard solutions of bovine serum albumin dissolved in 50 mM Tris-HCl (pH 8.2).

Example 5. The Activity of Rubisco and the Activity of PRK in Cell Extracts

(103) In order to study a possible requirement of heterologous chaperones of Rubisco in *S. cerevisiae*, the form-II Rubisco-encoding cbbM gene from *T. denitrificans* was codon-optimised and expressed from a centromeric vector, both alone and in combination with expression cassettes for the codon-optimised *E. coli* groEL/groES and/or *T. denitrificans* cbbO2/cbbQ2 genes. Analysis of ribulose-1,5-biphosphate-dependent CO_2 fixation by yeast cell extracts demonstrated that functional expression of *T. denitrificans* Rubisco in *S. cerevisiae* was observed upon co-expression of *E. coli* GroEL/GroES. Rubisco activity increased from <0.2 nmol·min^{-1}·(mg protein)$^{-1}$ to more than 6 nmol·min^{-1}·(mg protein)$^{-1}$. Results of these experiments are visualised in FIG. **1**, showing specific ribulose-1,5-bisphosphate carboxylase (Rubisco) activity in cell extracts of *S. cerevisiae* expressing Rubisco form II CbbM from *T. denitrificans* either alone (IMC033) or in combination with the *E. coli* chaperones GroEL/GroES (IMC035), The *T. denitrificans* chaperones CbbO2/CbbQ2 [20] (IMC034) or all four chaperones (IMC014). Heterologously expressed genes were codon optimised for expression in yeast and expressed from a single centromeric vector. Biomass samples were taken from anaerobic batch cultures on synthetic media (pH 5.0, 30° C.), sparged with nitrogen and containing 20 g 1-1 glucose as carbon source. Rubisco activities, measured as14CO2-fixation in cell extracts, in a wild-type reference strain and in *S. cerevisiae* strains expressing cbbM and cbbM-cbbQ2-cbbO2 were below the detection limit of the enzyme assay (0.2 nmol CO2 min−1 mg protein−1

(104) Co-expression of CbbO2/cbbQ2 did not result in a significant further increase of Rubisco activity. The positive effect of GroEL/GroES on Rubisco expression in *S. cerevisiae* demonstrates the potential value of this approach for metabolic engineering, especially when prokaryotic enzymes need to be functionally expressed in the cytosol of eukaryotes.

(105) The Spinach *oleracea* PRK gene was integrated together with *E. coli* groEL/groES and *T. denitrificans* cbbO2/cbbQ2 into the *S. cerevisiae* genome at the CAN1 locus, under control of the galactose-inducible GAL1 promoter. This induced in high PRK activities in cell extracts of *S. cerevisiae* strain IMU033, which additionally carried the centromeric expression cassette for *T. denitrificans* Rubisco. This engineered yeast strain was used to quantitatively analyze the physiological impacts of the expression of Rubisco and PRK.

(106) TABLE-US-00008 TABLE 8 IMU032 IMU033 (expressing (reference strain) PRK and Rubisco) CO_2 in inlet gas (%) 0 10 0 10 CO_2 in outlet gas (%) 0.89 ± 10.8 ± 1.02 ± 10.8 ± 0.03 0.0 0.00 0.1 Phosphoribulokinase 0.58 ± 0.51 ± 14.4 ± 15.2 ± (μmol mg protein^{-1} 0.09 0.12 1.5 1.0 min^{-1}) Rubisco (nmol mg <0.2* <0.2 4.59 ± 2.67 ± protein^{-1} min^{-1}) 0.30 0.28 Biomass yield on sugar 0.083 ± 0.084 ± 0.093 ± 0.095 ± (g g^{-1}) 0.000^{a} 0.000^{b} 0.001^{a} 0.000^{b} Ethanol yield on sugar 1.56 ± 1.56 ± 1.73 ± 1.73 ± (mol

mol.sup.−1) 0.03.sup.c 0.02.sup.d 0.02.sup.c 0.01.sup.d Glycerol yield on sugar 0.14 ± 0.12 ± 0.04 ± 0.01 ± (mol mol.sup.−1) 0.00.sup.e 0.00.sup.f 0.00.sup.e, g 0.00.sup.f, g

(107) Table 8 show increased ethanol yields on sugar of an *S. cerevisiae* strain expressing phosphoribulokinase (PRK) and Rubisco. Physiological analysis of *S. cerevisiae* IMU033 expressing PRK and Rubisco and the isogenic reference strain IMU032 in anaerobic chemostat cultures, grown at a dilution rate of 0.05 h−1 on a synthetic medium (pH 5) supplemented with 12.5 g 1-1 glucose and 12.5 g 1-1 galactose as carbon sources. To assess the impact of CO2 concentration, chemostat cultures were run sparged either with pure nitrogen gas or with a blend of 10% CO2 and 90% nitrogen. Results are represented as average ±mean deviations of data from independent duplicate chemostat experiments. Data pairs labelled with the same subscripts (a,a, b,b, etc.) are considered statistically different in a standard t-test (p<0.02).

(108) Expression of Rubisco and the four chaperones without co-expression of PRK (strain IMC014) did not result in decreased glycerol yield (0.13 mol mol.sup.−1) compared to the reference strain IMU032 (0.12 mol mol.sup.−1) in carbon-limited chemostat cultures supplemented with CO.sub.2, indicating that expression of a phosphoribulokinase (PRK) gene is required for the functional pathway in *S. cerevisiae* to decrease glycerol production. The physiological impact of expression of PRK and Rubisco on growth, substrate consumption and product formation in galactose-grown anaerobic batch cultures of *S. cerevisiae* was also investigated and compared with an isogenic reference strain. Growth conditions: T=30° C., pH 5.0, 10% CO.sub.2 in inlet gas. Two independent replicate experiments were carried out, whose growth kinetic parameters differed by less than 5%. Ethanol yield on galactose was 8% higher and glycerol production was reduced by 60% in the yeast cell in which PRK and Rubisco were functionally expressed, compared to the yeast cell lacking these enzymes. The differences were statistically significant (standard t-test (p value <0.02). The activities of phosphoribulokinase and of Rubisco in cell extracts of the engineered strain IMU033 (table 7) enable the use of CO.sub.2 as an electron acceptor. The ethanol yields and glycerol yields of strain IMU033 relative to the reference strain IMU032 (table 8) show that this is possible in an anaerobic fermentation with increased ethanol production.

(109) TABLE-US-00009 SEQUENCES Rubisco cbbM gene (synthetic; based on cbbM gene from *thiobacillus denitrificans*-pBTWW002, codon optimized Source: Hernandez et al 1996, GenBank ID: L37437.2) SEQ ID NO: 1
ATGGATCAATCTGCAAGATATGCTGACTTGTCTTTAAAGGAAGAAGATTTGATTAAAGGTG
GTAGACATATTTTGGTTGCTTACAAAATGAAACCAAAATCTGGTTATGGTTATTTGGAAGC
TGCTGCTCATTTTGCTGCTGAATCTTCTACAGGTACAAATGTTGAAGTTTCTACTACAGATG
ATTTTACAAAAGGTGTTGATGCTTTAGTTTACTACATCGATGAAGCTTCAGAAGATATGAG
AATTGCTTATCCATTGGAATTATTCGACAGAAATGTTACTGACGGAAGATTCATGTTAGTTT
CTTTTTTGACTTTGGCTATTGGTAACAATCAAGGAATGGGAGATATAGAACATGCAAAAAT
GATAGATTTTTACGTTCCAGAAAGATGTATTCAAATGTTTGATGGTcCAGCTACAGATATTT
CTAATTTGTGGAGAATTTTGGGTAGACCAGTAGTTAATGGTGGTTATATTGCTGGTACTAT
TATTAAGCCAAAATTGGGTTTAAGACCAGAAcCATTTGCTAAAGCTGCTTATCAATTTTGG
TTGGGTGGAGATTTTATCAAGAATGACGAACCACAAGGTAATCAAGTTTTTTGTCCATTGA
AAAAAGTTTTGCCATTGGTTTACGATGCTATGAAAAGAGCACAAGATGATACTGGTCAAGC
AAAATTGTTTTCTATGAATATTACTGCAGACGATCATTATGAAATGTGTGCAAGAGCTGAT
TATGCTTTGGAAGTTTTCGGTCCAGATGCAGATAAATTGGCTTTTTTGGTAGATGGTTACG
TTGGAGGTCCAGGAATGGTTACTACTGCTAGAAGGCAATATCCTGGTCAATATTTGCATTA
TCATAGAGCAGGTCACGGTGCTGTTACTTCTCCATCTGCTAAAAGAGGTTATACTGCTTTT
GTTTTGGCTAAAATGTCTAGATTGCAAGGCGCTTCAGGTATTCATGTTGGTACTATGGGTT
ATGGAAAAATGGAAGGAGAAGGCGACGATAAGATTATTGCTTATATGATAGAAAGGGACG
AATGTCAAGGTCCAGTTTATTTTCAAAAATGGTACGGTATGAAACCAACTACTCCAATTAT
CTCCGGAGGAATGAATGCTTTGAGATTGCCTGGTTTTTTCGAAAATTTGGGTCATGGTAAC
GTTATTAATACTGCAGGTGGTGGTTCTTACGGTCATATTGATTCTCCTGCTGCTGGTGCTA

TTTCTTTGAGACAATCTTACGAATGTTGGAAACAAGGTGCAGATCCAATTGAATTTGCTAA GGAACATAAGGAATTTGCAAGAGCTTTTGAATCTTTTCCAAAAGATGCTGATAAGTTATTT CCAGGATGGAGAGAAAAATTGGGAGTTCATTCTTAA Translated protein sequence of cbbM gene from *Thiobacillus denitrificans* SEQ ID NO: 2 MDQSARYADLSLKEEDLIKGGRHILVAYKMKPKSGYGYLEAAAHFAAESSTGTNVEVSTTD DFTKGVDALVYYIDEASEDMRIAYPLELFDRNVTDGRFMLVSFLTLAIGNNQGMGDIEHAK MIDFYVPERCIQMFDGPATDISNLWRILGRPVVNGGYIAGTIIKPKLGLRPEPFAKAAYQFWL GGDFIKNDEPQGNQVFCPLKKVLPLVYDAMKRAQDDTGQAKLFSMNITADDHYEMCARAD YALEVFGPDADKLAFLVDGYVGGPGMVTTARRQYPGQYLHYHRAGHGAVTSPSAKRGYTA FVLAKMSRLQGASGIHVGTMGYGKMEGEGDDKIIAYMIERDECQGPVYFQKWYGMKPTTP IISGGMNALRLPGFFENLGHGNVINTAGGGSYGHIDSPAAGAISLRQSYECWKQGADPIEFA KEHKEFARAFESFPKDADKLFPGWREKLGVHS prk gene from *Spinacea oleracea*-pBTWW001, plasmid constructed using restriction and ligation. Source: Milanez and Mural 1988, GenBank ID: M21338.1 SEQ ID NO: 3 ATGTCACAACAACAAACAATTGTGATTGGTTTAGCAGCAGATTCAGGTTGTGGTAAGAGTA CATTCATGAGGAGGTTAACAAGTGTTTTCGGTGGCGCGGCCGAGCCACCAAAGGGTGGTA ACCCAGATTCAAACACATTGATTAGTGACACTACTACTGTTATCTGTTTGGATGATTTTCAT TCCCTTGATAGAAATGGCAGGAAAGTGGAAAAAGTTACTGCTTTAGACCCAAAAGCTAATG ATTTTGATCTTATGTATGAACAAGTTAAGGCTTTGAAAGAAGGTAAAGCTGTTGATAAACC TATTTATAATCATGTTTCTGGTTTGTTGGACCCTCCTGAGCTTATTCAACCTCCTAAGATCT TGGTCATTGAAGGGTTACACCCCATGTATGACGCACGTGTGAGGGAATTGCTAGACTTCAG CATCTACTTGGACATTAGCAATGAAGTTAAATTTGCCTGGAAAATTCAGAGAGACATGAAA GAAAGAGGACACAGTCTTGAAAGCATCAAAGCCAGTATTGAATCCAGAAAGCCAGATTTTG ATGCTTACATTGACCCACAAAAGCAGCATGCTGATGTAGTGATTGAAGTATTGCCAACTGA ACTCATTCCTGATGATGATGAAGGCAAAGTGTTGAGAGTAAGGATGATTCAGAAAGAAGG AGTCAAGTTTTTCAACCCAGTTTACTTGTTTGATGAAGGATCTACCATTTCATGGATTCCAT GTGGTAGAAAATTAACATGTTCTTACCCTGGTATCAAATTTTCCTATGGCCCAGACACCTT CTATGGCAACGAGGTGACAGTAGTAGAGATGGATGGGATGTTTGACAGATTAGACGAACT AATCTACGTCGAAAGCCATTTGAGCAATCTATCAACCAAGTTTTATGGTGAAGTCACTCAA CAAATGTTGAAGCACCAAAATTTCCCAGGAAGCAACAATGGAACTGGTTTCTTCCAAACCA TAATTGGATTGAAGATCAGAGACTTGTTCGAGCAGCTCGTTGCTAGCAGGTCTACAGCAAC TGCAACAGCTGCTAAAGCCTAG Translated protein sequence of prk gene from *Spinacea oleracea* SEQ ID NO: 4 MSQQQTIVIGLAADSGCGKSTFMRRLTSVFGGAAEPPKGGNPDSNTLISDTTTVICLDDFHS LDRNGRKVEKVTALDPKANDFDLMYEQVKALKEGKAVDKPIYNHVSGLLDPPELIQPPKIL VIEGLHPMYDARVRELLDFSIYLDISNEVKFAWKIQRDMKERGHSLESIKASIESRKPDFDA YIDPQKQHADVVIEVLPTELIPDDDEGKVLRVRMIQKEGVKFFNPVYLFDEGSTISWIPCGR KLTCSYPGIKFSYGPDTFYGNEVTVVEMDGMFDRLDELIYVESHLSNLSTKFYGEVTQQML KHQNFPGSNNGTGFFQTIIGLKIRDLFEQLVASRSTATATAAKA SEQ ID NO: 5 cbbQ2 gene (synthetic, based on cbbQ2 gene from *Thiobacillus denitrificans*-codon optimized, original sequence obtained from Beller et al 2006, GenBank Gene ID: 3672366, Protein ID: AAZ98590.1 ATGACTACTAACAAGGAACAATACAAGGTTCACCAAGAACCATACTACCAAGCTCAAGGTA GAGAAGTTCAATTGTACGAAGCTGCTTACAGAAACAGATTGCCAGTTATGGTTAAGGGTCC AACTGGTTGTGGTAAGTCTAGATTCGTTGAATACATGGCTTGGAAGTTGAACAAGCCATTG ATCACTGTTGCTTGTAACGAAGACATGACTGCTTCTGACTTGGTTGGTAGATACTTGTTGG AAGCTAACGGTACTAGATGGTTGGACGGTCCATTGACTACTGCTGCTAGAATCGGTGCTAT CTGTTACTTGGACGAAGTTGTTGAAGCTAGACAAGACACTACTGTTGTTATCCACCCATTG ACTGACCACAGAAGAACTTTGCCATTGGACAAGAAGGGTGAATTGATCGAAGCTCACCCA GACTTCCAATTGGTTATCTCTTACAACCCAGGTTACCAATCTTTGATGAAGGACTTGAAGC

AATCTACTAAGCAAAGATTCGCTGCTTTCGACTTCGACTACCCAGACGCTGCTTTGGAAAC TACTATCTTGGCTAGAGAAACTGGTTTGGACGAAACTACTGCTGGTAGATTGGTTAAGATC GGTGGTGTTGCTAGAAACTTGAAGGGTCACGGTTTGGACGAAGGTATCTCTACTAGATTGT TGGTTTACGCTGCTACTTTGATGAAGGACGGTGTTGACGCTGGTGACGCTTGTAGAATGGC TTTGGTTAGACCAATCACTGACGACGCTGACATCAGAGAAACTTTGGACCACGCTATCGAC GCTACTTTCGCTTAA Translated protein sequence of cbbQ2 gene from *Thiobacillus denitrificans* SEQ ID NO: 6

MTTNKEQYKVHQEPYYQAQGREVQLYEAAYRNRLPVMVKGPTGCGKSRFVEYMAWKLNK PLITVACNEDMTASDLVGRYLLEANGTRWLDGPLTTAARIGAICYLDEVVEARQDTTVVIHP LTDHRRTLPLDKKGELIEAHPDFQLVISYNPGYQSLMKDLKQSTKQRFAAFDFDYPDAALE TTILARETGLDETTAGRLVKIGGVARNLKGHGLDEGISTRLLVYAATLMKDGVDAGDACRM ALVRPITDDADIRETLDHAIDATFA cbbO2 gene (Synthetic, based on cbbO2 gene from *Thiobacillus denitrificans*-codon optimized, original sequence obtained from Beller et al 2006, GenBank Gene ID: 3672365, Protein ID: YP_316394.1 SEQ ID NO: 7

ATGGCTGCTTACTGGAAGGCTTTGGACACTAGATTCGCTCAAGTTGAAGAAGTTTTCGACG ACTGTATGGCTGAAGCTTTGACTGTTTTGTCTGCTGAAGGTGTTGCTGCTTACTTGGAAGC TGGTAGAGTTATCGGTAAGTTGGGTAGAGGTGTTGAACCAATGTTGGCTTTCTTGGAAGAA TGGCCATCTACTGCTCAAGCTGTTGGTGAAGCTGCTTTGCCAATGGTTATGGCTTTGATCC AAAGAATGCAAAAGTCTCCAAACGGTAAGGCTATCGCTCCATTCTTGCAAACTTTGGCTCC AGTTGCTAGAAGATTGCAATCTGCTGAACAATTGCAACACTACGTTGACGTTACTTTGGAC TTCATGACTAGAACTACTGGTTCTATCCACGGTCACCACACTACTTTCCCATCTCCAGGTTT GCCAGAATTCTTCGCTCAAGCTCCAAACTTGTTGAACCAATTGACTTTGGCTGGTTTGAGA AACTGGGTTGAATACGGTATCAGAAACTACGGTACTCACCCAGAAAGACAACAAGACTACT TCTCTTTGCAATCTGCTGACGCTAGAGCTGTTTTGCAAAGAGAAAGACACGGTACTTTGTT GGTTGACGTTGAAAGAAAGTTGGACTTGTACTTGAGAGGTTTGTGGCAAGACCACGACCA CTTGGTTCCATACTCTACTGCTTTCGACGAAATCAGAAAGCCAGTTCCATACTACGACAAG TTGGGTATGAGATTGCCAGACGTTTACGACGACTTGGTTTTGCCATGTCCAGCTGGTAGAG GTGGTGCTGGTGGTGAAGACGTTTTGGTTTCTGGTTTGGACAGATACAGAGCTACTTTGGC TCACATGGTTGGTCACAGAAGATGGTCTGAAGCTCAAATCGCTGACAACTGGTCTCCATTC CAAAGAATGGCTGTTGAATTCTTCGAAGACTGTAGAGTTGAAACTTTGTTGATGAGAGAAT ACCCAGGTTTGGCTAGAATCTTCAGAGCTTTGCACCCAAAGCCAGTTGAAGCTGCTTGTGA CGGTGAAACTACTTCTTGTTTGAGACACAGATTGGCTATGTTGTCTAGAGCTTTCATCGAC CCAGACCACGGTTACGCTGCTCCAGTTTTGAACGACTTCGTTGCTAGATTCCACGCTAGAT TGGCTGACGGTACTTCTTCTACTTCTGAAATGGCTGACTTGGCTTTGTCTTACGTTGCTAA GACTAGAAGACCATCTGACCAATTCGCTAAGGTTCACTTCGACGACACTGTTGTTGACTAC AGAGACGACAACAGACAATTGTGGAAGTTCATCGAAGAAGGTGACGAAGAAGAAGCTTTC GACGCTAAGAGAAAGATCGAACCAGGTGAAGAAATCCAAGGTTTGCCACCAAGACACTAC CCAGAATGGGACTACACTTCTCAAACTTACAGACCAGACTGGGTTTCTGTTTACGAAGGTT TGCACAGATCTGGTAACGCTGGTGACATCGACAGATTGTTGGCTAAGCACGCTGCTTTGGC TAAGAGATTGAAGAAGATGTTGGACTTGTTGAAGCCACAAGACAAGGTTAGAGTTAGATAC CAAGAAGAAGGTTCTGAATTGGACTTGGACGTTGCTATCAGATCTTTGATCGACTTCAAGG GTGGTGCTACTCCAGACCCAAGAATCAACATGTCTCACAGATCTGACGGTAGAGACATCGC TGTTATGTTGTTGTTGGACTTGTCTGAATCTTTGAACGAAAAGGCTGCTGGTGCTGGTCAA ACTATCTTGGAATTGTCTCAAGAAGCTGTTTCTTTGTTGGCTTGGTCTATCGAAAAGTTGG GTGACCCATTCGCTATCGCTGGTTTCCACTCTAACACTAGACACGACGTTAGATACTTCCA CATCAAGGGTTACTCTGAAAGATGGAACGACGACGTTAAGGCTAGATTGGCTGCTATGGA AGCTGGTTACTCTACTAGAATGGGTGCTGCTATGAGACACGCTGCTCACTACTTGTCTGCT AGACCAGCTGACAAGAAGTTGATGTTGATCTTGACTGACGGTAGACCATCTGACGTTGACG CTGCTGACGAAAGATTGTTGGTTGAAGACGCTAGACAAGCTGTTAAGGAATTGGACAGAC

AAGGTATCTTCGCTTACTGTATCTCTTTGGACGCTCAATTGAAGGCTGGTGCTGACGACTA CGTTGCTGAAATCTTCGGTAGACAATACACTGTTATCGACAGAGTTGAAAGATTGCCAGAA AGATTGCCAGAATTGTTCATGGCTTTGACTAAGTAA Translated protein sequence of cbbO2 gene from *Thiobacillus denitrificans* SEQ ID NO: 8 MAAYWKALDTRFAQVEEVFDDCMAEALTVLSAEGVAAYLEAGRVIGKLGRGVEPMLAFLE EWPSTAQAVGEAALPMVMALIQRMQKSPNGKAIAPFLQTLAPVARRLQSAEQLQHYVDVTL DFMTRTTGSIHGHHTTFPSPGLPEFFAQAPNLLNQLTLAGLRNWVEYGIRNYGTHPERQQD YFSLQSADARAVLQRERHGTLLVDVERKLDLYLRGLWQDHDHLVPYSTAFDEIRKPVPYYD KLGMRLPDVYDDLVLPCPAGRGGAGGEDVLVSGLDRYRATLAHMVGHRRWSEAQIADNW SPFQRMAVEFFEDCRVETLLMREYPGLARIFRALHPKPVEAACDGETTSCLRHRLAMLSRA FIDPDHGYAAPVLNDFVARFHARLADGTSSTSEMADLALSYVAKTRRPSDQFAKVHFDDTV VDYRDDNRQLWKFIEEGDEEEAFDAKRKIEPGEEIQGLPPRHYPEWDYTSQTYRPDWVSVY EGLHRSGNAGDIDRLLAKHAALAKRLKKMLDLLKPQDKVRVRYQEEGSELDLDVAIRSLID FKGGATPDPRINMSHRSDGRDIAVMLLLDLSESLNEKAAGAGQTILELSQEAVSLLAWSIEK LGDPFAIAGFHSNTRHDVRYFHIKGYSERWNDDVKARLAAMEAGYSTRMGAAMRHAAHYL SARPADKKLMLILTDGRPSDVDAADERLLVEDARQAVKELDRQGIFAYCISLDAQLKAGAD DYVAEIFGRQYTVIDRVERLPERLPELFMALTK GroEL gene (synthetic, based on GroEL from *E. coli*-codon optimized, original sequence obtained from Durfee et al 2008, Gene ID: 6061450, Protein ID: YP_001732912.1 SEQ ID NO: 9 ATGGCTGCTAAGGACGTTAAGTTCGGTAACGACGCTAGAGTTAAGATGTTGAGAGGTGTT AACGTTTTGGCTGACGCTGTTAAGGTTACTTTGGGTCCAAAGGGTAGAAACGTTGTTTTGG ACAAGTCTTTCGGTGCTCCAACTATCACTAAGGAGGGTGTTTCTGTTGCTAGAGAAATCGA ATTGGAAGACAAGTTCGAAAACATGGGTGCTCAAATGGTTAAGGAAGTTGCTTCTAAGGCT AACGACGCTGCTGGTGACGGTACTACTACTGCTACTGTTTTGGCTCAAGCTATCATCACTG AACGACGCTGCTGGTGACGGTACTACTACTGCTACTGTTTTGGCTCAAGCTATCATCACTG AAGGTTTGAAGGCTGTTGCTGCTGGTATGAACCCAATGGACTTGAAGAGAGGTATCGACA AGGCTGTTACTGCTGCTGTTGAAGAATTGAAGGCTTTGTCTGTTCCATGTTCTGACTCTAA GGCTATGGCTCAAGTTGGTACTATCTCTGCTAAGTCTGACGAAACTGTTGGTAAGTTGATC GCTGAAGCTATGGACAAGGTTGGTAAGGAAGGTGTTATCACTGTTGAAGACGGTACTGGT TTGCAAGACGAATTGGAGGTTGTTGAAGGTATGGAATTCGACAGAGGTTACTTGTGTCCAT ACTTCATCAACAAGCCAGAAACTGGTGCTGTTGAATTGGAATCTCCATTCATCTTGTTGGC TGACAAGAAGATCTCTAACATCAGAGAAATGTTGGCAGTTTTGGAAGCTGTTGCTAAGGCT GGTAAGCCATTGTTGATCATCGCTGAAGACGTTGAAGGTGAAGCTTTGGCTACTTTGGTTG TTAACACTATGAGAGGTATCGTTAAGGTTGCTGCTGTTAAGGGTCCAGGTTTCGGTGAGAG AAGAAAGGCTATGTTGCAAGACATCGCTACTTTGACTGGTGGTACTGTTATCTCTGAAGAA ATCGGTATGGAATTGGAAAAGGGTACTTTGGAAGACTTGGGTCAAGCTAAGAGAGTTGTTA TCAACAAGGACACTACTACTATCATCGACGGTGTTGGTGAAGAAGCTGCTATCCAAGGTAG AGTTGCTCAAATCAGACAACAAATCGAAGAAGCTACTTCTGACTACGACAGAGAAAAGTTG CAAGAAAGAGTTGCTAAGTTGGCTGGTGGTGTTGGTGTTATGAAGGTTGGTGCTGCTACTG AAGTTGAAATGAAGGAAAAGAAGGCTAGAGTTGAAGACGCTTTGCACGCTACTAGAGCTG CTGTTGAAGAAGGTGTTGTTGCTGGTGGTGGTGTTGCTTTGATCAGAGTTGCTTCTAAGTT GGCTGACTTGAGAGGTCAAAACGAAGACCAAAACGTTGGTATCAAGGTTGCTTTGAGAGC TATGGAAGCTCGATTGAGACAAATCGTTTTGAACTGTGGTGAAGAACCATCTGTTGTTGCT AACACTGTTAAGGGTGGTGACGGTAACTACGGTTACAACGCTGCTACTGAAGAATACGGT AACATGATCGACATGGGTATCTTGGACCCAAGTAAGGTTACTAGATCTGCTTTGCAATAGG CTGCTTCTGTTGCTGGTTTGATGATCACTACTGAATGTATGGTTACTGACTTGCCAAAGAA CGACGCTGCTGACTTGGGTGCTGCTGGTGGTATGGGTGGTATGGGTGGTATGGGTGGTAT GATGTAA Translated protein sequence of GroEL gene from *E. coli* SEQ ID NO: 10 MAAKDVKFGNDARVKMLRGVNVLADAVKVTLGPKGRNVVLDKSFGAPTITKDGVSVAREI

ELEDKFENMGAQMVKEVASKANDAAGDGTTTATVLAQAIITEGLKAVAAGMNPMDLKRGI DKAVTAAVEELKALSVPCSDSKAIAQVGTISANSDETVGKLIAEAMDKVGKEGVITVEDGTG LQDELDVVEGMQFDRGYLSPYFINKPETGAVELESPFILLADKKISNIREMLPVLEAVAKAG KPLLIIAEDVEGEALATLVVNTMRGIVKVAAVKAPGFGDRRKAMLQDIATLTGGTVISEEIG MELEKATLEDLGQAKRVVINKDTTTIIDGVGEEAAIQGRVAQIRQQIEEATSDYDREKLQER VAKLAGGVAVIKVGAATEVEMKEKKARVEDALHATRAAVEEGVVAGGGVALIRVASKLADL RGQNEDQNVGIKVALRAMEAPLRQIVLNCGEEPSVVANTVKGGDGNYGYNAATEEYGNMI DMGILDPTKVTRSALQYAASVAGLMITTECMVTDLPKNDAADLGAAGGMGGMGGMGGMM GroES gene (synthetic, based on GroES *E. coli*-codon optimized, original sequence obtained from Durfee et al 2008, Gene ID: 6061370, Protein ID: YP_001732911.1 SEQ ID NO: 11
ATGAACATCAGACCATTGCACGACAGAGTTATCGTTAAGAGAAAGGAAGTTGAAACTAAGT CTGCTGGTGGTATCGTTTTGACTGGTTCTGCTGCTGCTAAGTCTACTAGAGGTGAAGTTTT GGCTGTTGGTAACGGTAGAATCTTGGAAAACGGTGAAGTTAAGCCATTGGACGTTAAGGT TGGTGACATCGTTATCTTCAACGACGGTTACGGTGTTAAGTCTGAAAAGATCGACAACGAA GAAGTTTTGATCATGTCTGAATCTGACATCTTGGCTATCGTTGAAGCTTAA Translated protein sequence of GroES gene from *E. coli* SEQ ID NO: 12
MNIRPLHDRVIVKRKEVETKSAGGIVLTGSAAAKSTRGEVLAVGNGRILENGEVKPLDVKV GDIVIFNDGYGVKSEKIDNEEVLIMSESDILAIVEA

Claims

1. A recombinant yeast cell comprising: one or more recombinant heterologous, nucleic acid sequences encoding Ribulose-1,5-bisphosphate carboxylase oxygenase (Rubisco) and phosphoribulokinase (PRK), wherein the amino acid sequence of the Rubisco has at least 80% sequence identity with the sequence of SEQ ID NO: 2, and the amino acid sequence of the PRK has at least 70% sequence identity with the sequence of SEO ID NO: 4; and nucleic acid sequences encoding a GroEL or a functional homologue of GroEL, and a GroES or a functional homologue of GroES, wherein the respective GroEL and GroES homologues are functional as molecular chaperones, wherein the amino acid sequence of the GroEL or the functional homologue of GroEL has at least 70% sequence identity with the sequence of SEO ID NO: 10; and wherein the amino acid sequence of the GroES or the functional homologue of GroES has at least 70% sequence identity with the sequence of SEO ID NO: 12.
2. The recombinant yeast cell of claim 1, wherein the Rubisco uses CO2 as an electron acceptor.
3. The recombinant yeast cell of claim 1, wherein the Rubisco is a single subunit Rubisco.
4. The recombinant yeast cell of claim 1, wherein the Rubisco is a prokaryotic form-II Rubisco.
5. The recombinant yeast cell of claim 1, wherein the PRK is a PRK originating from a eukaryote.
6. The recombinant yeast cell of claim 5, wherein the PRK originates from a Caryophyllales plant.
7. The recombinant yeast cell of claim 6, wherein the Caryophyllales plant is Amaranthaceae or *Spinacia.*
8. The recombinant yeast cell of claim 1, wherein the genus of said yeast cell is selected from the group consisting of Saccharomycesaceae, *Schizosaccharomyces, Torulaspora, Kluyveromyces, Pichia, Zygosaccharomyces, Brettanomyces, Metschnikowia, Issatchenkia, Kloeckera*, and *Aureobasidium.*
9. The recombinant yeast cell of claim 8, wherein the genus of yeast cell is Saccharomycesceae.
10. The recombinant yeast cell of claim 9, wherein the yeast cell is selected from the group consisting of *Saccharomyces cerevisiae, Saccharomyces pastorianus*, *Saccharomyces beticus, Saccharomyces fermentati, Saccharomyces paradoxus, Saccharomyces uvarum*, and *Saccharomyces bayanus.*
11. A method for preparing an alcohol under anaerobic conditions comprising fermenting a carbon

source and the recombinant yeast cell of claim 1.
12. The recombinant yeast cell of claim 7, wherein the PRK is a *Spinacia* PRK.
13. The recombinant yeast cell of claim 9, wherein the yeast cell is a *Saccharomyces cerevisiae.*
14. The recombinant yeast cell of claim 1, wherein the amino acid sequence of the GroEL or the functional homologue of GroEL has at least 75% sequence identity with the sequence of SEQ ID NO: 10; and the amino acid sequence of the GroES or the functional homologue of GroES has at least 75% sequence identity with the sequence of SEQ ID NO: 12.
15. The recombinant yeast cell of claim 1, wherein the amino acid sequence of the GroEL or the functional homologue of GroEL has at least 80% sequence identity with the sequence of SEQ ID NO: 10; and the amino acid sequence of the GroES or the functional homologue of GroES has at least 80% sequence identity with the sequence of SEQ ID NO: 12.
16. The recombinant yeast cell of claim 1, further comprising a heterologous nucleic acid sequence encoding a dehydrogenase.
17. The recombinant yeast cell of claim 16, wherein the dehydrogenase is a xylitol dehydrogenase.
18. The recombinant yeast cell of claim 1, wherein the amino acid sequence of the GroEL or the functional homologue of GroEL has at least 85% sequence identity with the sequence of SEQ ID NO: 10; and the amino acid sequence of the GroES or the functional homologue of GroES has at least 85% sequence identity with the sequence of SEQ ID NO: 12.
19. The recombinant yeast cell of claim 1, wherein the amino acid sequence of the GroEL or the functional homologue of GroEL has at least 90% sequence identity with the sequence of SEQ ID NO: 10; and the amino acid sequence of the GroES or the functional homologue of GroES has at least 90% sequence identity with the sequence of SEQ ID NO: 12.
20. The recombinant yeast cell of claim 1, wherein the amino acid sequence of the GroEL or the functional homologue of GroEL has at least 95% sequence identity with the sequence of SEQ ID NO: 10; and the amino acid sequence of the GroES or the functional homologue of GroES has at least 95% sequence identity with the sequence of SEQ ID NO: 12.
21. The recombinant yeast cell of claim 1, wherein the amino acid sequence of the Rubisco has at least 85% sequence identity with the sequence of SEQ ID NO: 2.
22. The recombinant yeast cell of claim 1, wherein the amino acid sequence of the Rubisco has at least 90% sequence identity with the sequence of SEQ ID NO: 2.
23. The recombinant yeast cell of claim 1, wherein the amino acid sequence of the Rubisco has at least 95% sequence identity with the sequence of SEQ ID NO: 2.
24. The recombinant yeast cell of claim 1, wherein the amino acid sequence of the PRK has at least 75% sequence identity with the sequence of SEQ ID NO: 4.
25. The recombinant yeast cell of claim 1, wherein the amino acid sequence of the PRK has at least 80% sequence identity with the sequence of SEQ ID NO: 4.
26. The recombinant yeast cell of claim 1, wherein the amino acid sequence of the PRK has at least 85% sequence identity with the sequence of SEQ ID NO: 4.
27. The recombinant yeast cell of claim 1, wherein the amino acid sequence of the PRK has at least 90% sequence identity with the sequence of SEQ ID NO: 4.
28. The recombinant yeast cell of claim 1, wherein the amino acid sequence of the PRK has at least 95% sequence identity with the sequence of SEQ ID NO: 4.